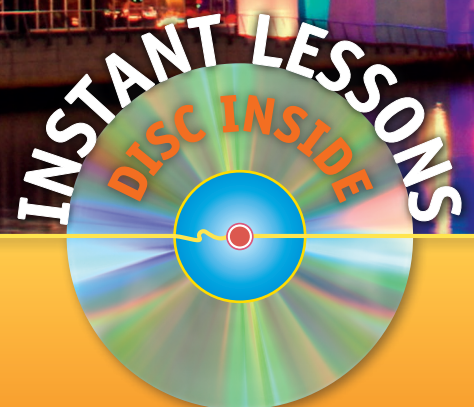




MATHS in Sustainability

Christine Henderson



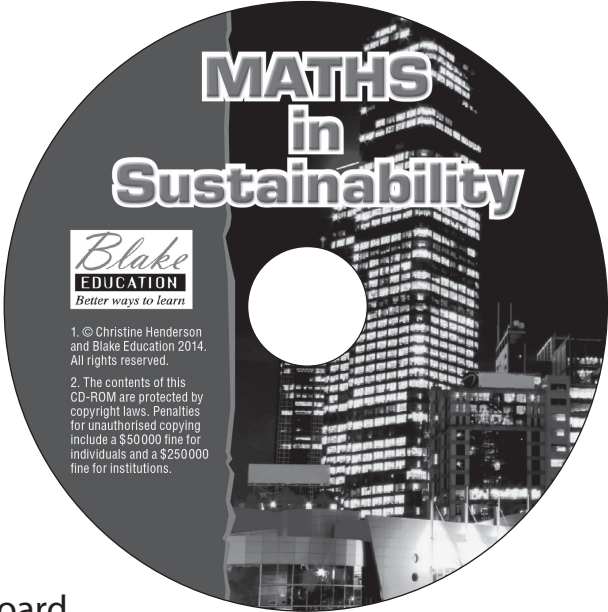
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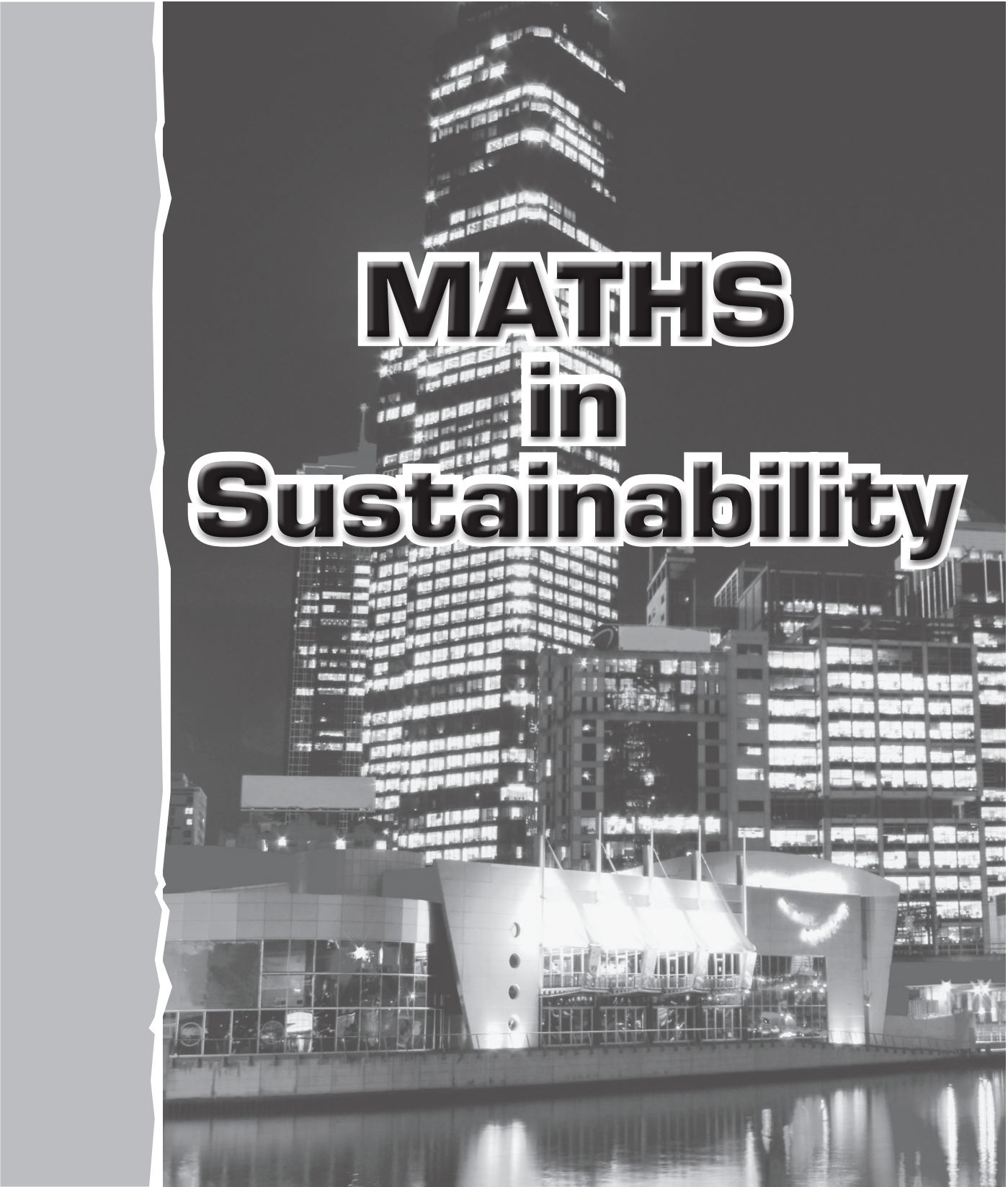


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Australian curriculum alignment	Year	Units
ACMNA182 Use index notation with numbers to establish the index laws with positive integral indices and the zero index	8	21
ACMNA187 Solve problems involving the use of percentages, including percentage increases and decreases, with and without digital technologies	8	13, 16
ACMNA188 Solve a range of problems involving rates and ratios, with and without digital technologies	8	7
ACMNA193 Plot linear relationships on the Cartesian plane with and without the use of digital technologies	8	12
ACMNA194 Solve linear equations using algebraic and graphical techniques. Verify solutions by substitution	8	12
ACMNA208 Solve problems involving direct proportion. Explore the relationship between graphs and equations corresponding to simple rate problems	9	9, 14, 15
ACMNA209 Apply index laws to numerical expressions with integer indices	9	5
ACMNA210 Express numbers in scientific notation	9	5
ACMNA236 Solve linear inequalities and graph their solutions on a number line	10	11
ACMNA237 Solve linear simultaneous equations, using algebraic and graphical techniques including using digital technology	10	20
ACMNA239 Explore the connection between algebraic and graphical representations of relations such as simple quadratics, circles and exponentials using digital technology as appropriate	10	19



Australian curriculum alignment	Year	Units
ACMMG219 Investigate very small and very large time scales and intervals	9	21
ACMMG242 Solve problems involving surface area and volume for a range of prisms, cylinders and composite solids	10	3, 4
ACMMG245 Solve right-angled triangle problems including those involving direction and angles of elevation and depression	10	2
ACMSP169 Identify and investigate issues involving numerical data collected from primary and secondary sources	7	16
ACMSP170 Construct and compare a range of data displays including stem-and-leaf plots and dot plots	7	6
ACMSP171 Calculate mean, median, mode and range for sets of data. Interpret these statistics in the context of data	7	6
ACMSP172 Describe and interpret data displays using median, mean and range	7	10
ACMSP205 Describe events using language of 'at least', exclusive 'or' (A or B but not both), inclusive 'or' (A or B or both) and 'and'	8	17
ACMSP207 Investigate the effect of individual data values, including outliers, on the mean and median	8	10
ACMSP225 List all outcomes for two-step chance experiments, both with and without replacement using tree diagrams or arrays. Assign probabilities to outcomes and determine probabilities for events	9	8
ACMSP226 Calculate relative frequencies from given or collected data to estimate probabilities of events involving 'and' or 'or'	9	17
ACMSP292 Represent events in two-way tables and Venn diagrams and solve related problems	8	18



MATHS **in** **Sustainability**

Christine Henderson

The book was commissioned by Lisa Tancredi
Project management and typesetting by Piper Print Media
Cover design by Jo-Anne Ridgway/Polar Design

Christine Henderson has spent over twenty years as a maths and physics educator in secondary schools and universities in Australia, the UK and Africa. Christine writes maths and science scripts for a leading Australian educational film production company.

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ISBN 978 1 92571 066 3

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Maths in Sustainability is a collection of photocopiable student work sheets that feature the application of mathematical knowledge and skills in engaging, real-world contexts. Transport, energy sources, recycling, water, composting, electricity, food miles and the Earth in space are some of the topics used to provide fresh approaches for teaching Australian Curriculum content. The activities will help students recognise how mathematical skills can be used in daily life.

The work sheets are designed to be used in the classroom or for homework assignments across years 7 to 10. Links to the Australian Curriculum can be found in the alignment grid on the inside front cover of this text. All work sheets are included in the PDF version of the book on the enclosed CD-ROM.

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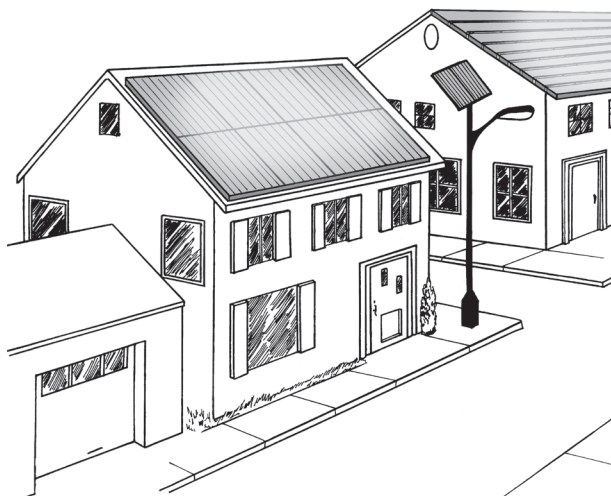
Answers 46



We use electrical energy to power our homes, schools and places of work. Generating electricity by burning coal or gas in a power station is not sustainable, and produces carbon dioxide. Using sunlight to make electricity is sustainable. A solar panel is a grid of photovoltaic cells that convert sunlight into electric power.

Power is measured in watts.
One kilowatt (kW) = 1000 watts (W).

A single solar panel produces between 100 and 200 watts, depending on the brand and size. A system of several panels can be installed on a house roof to generate some or all of the energy used by the household.



- 1 How many 100 W panels are needed to produce each amount of electrical power?
 - a 1 kW
 - b 3 kW

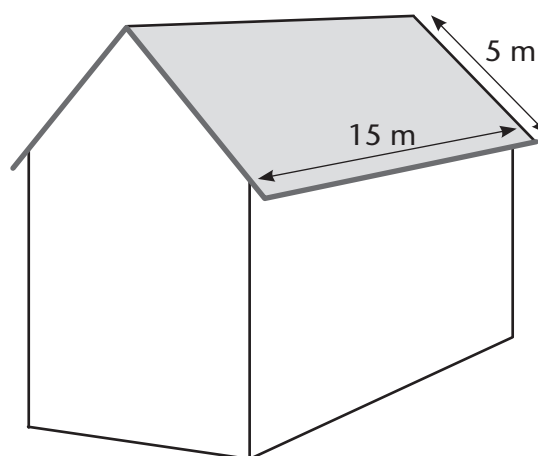
- 2 A solar panel system contains ten 200 watt panels.
What is the total power generated? Answer in kilowatts.

- 3 A 100 W solar panel measures 1670 mm long by 990 mm wide.
There are 1000 millimetres (mm) in one metre (m).
 - a What is the area of the panel in square millimetres?
 - b Converting to metres, the panel measures _____ m long by _____ m wide.
 - c What is the panel's area in square metres?
 - d How many square millimetres are there in one square metre?

- 4 A solar energy system installed on a school roof is made up of ten 200 W panels, each 1.8 m long and 1.1 m wide. The roof is a rectangle 30 m long and 8 m wide.
 - a What is the area of one solar panel in square metres?
 - b What area of the roof is covered by solar panels?
 - c What is the area of the roof?
 - d What percentage of the roof area is covered with solar panels?
 - e How many whole panels will fit across 30 metres of roof if each panel is 1.1 m wide and they are laid side-by-side?
 - f A maximum of four rows of panels can be fitted onto the roof.
How many panels will that be?

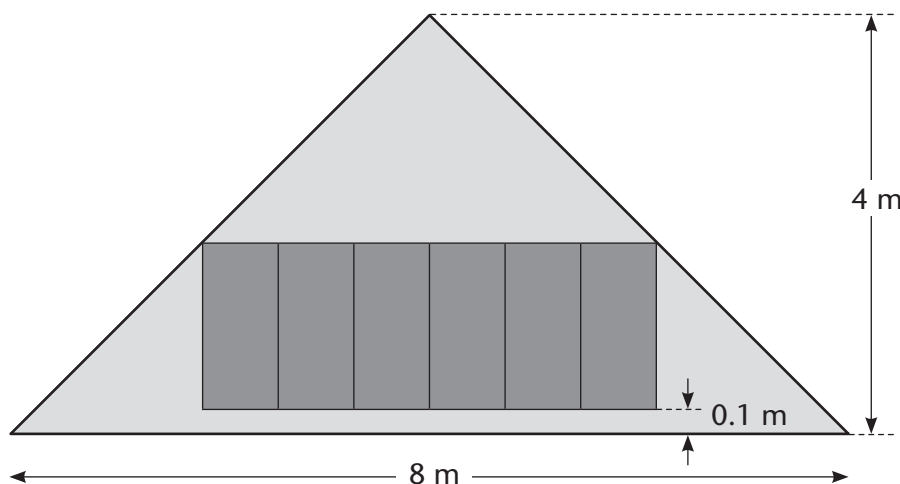


- 5** Sonny and Mia want to cover the north-facing roof of their new home with a solar energy system. The panels they have chosen are each 1.58 m long and 0.8 m wide. The roof measures 15 m by 5 m.



- If the panels are fitted with their longer side down the roof slope, how many panels will fit in a single row across the roof?
- How many rows of panels will fit on the roof?
- What is the maximum number of panels that will fit on the roof?
- Calculate the area of each panel.
- What is the total roof area covered by panels?
- What is the roof area left uncovered?

- 6** The north-facing section of the roof on this house is the shape of a 45 degrees isosceles triangle.



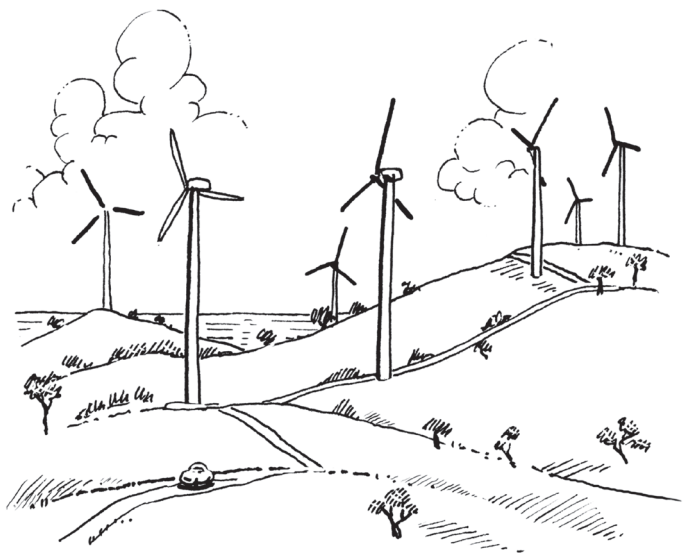
- What is the area of this section of the roof?
- What area is taken up by six solar panels, each 1.2 m long by 0.9 m wide?
- What percentage of this roof section is covered by panels?
- What is the height of the top triangle above the panels?
(Hint: use the length of one solar panel and the information shown in the diagram.)
- What is the length of the base of the top triangle?
- Use your answers to parts d and e to calculate the area of the top triangle.



Wind is an important source of sustainable energy. A wind turbine uses the wind to turn its large blades, which drive an electricity generator at the top of the tower. Electric cables carry the electricity to the mains grid.

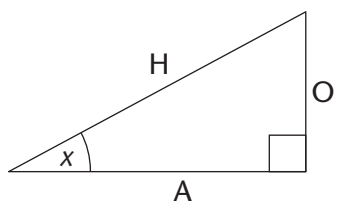
Wind farms range from a community wind farm with a few turbines, to large commercial farms of several hundred turbines.

Wind farms are sited in areas of high wind, such as upland ridges and the west coasts of Victoria and Tasmania, which lie in the 'Roaring Forties', a high wind zone.



- 1 A typical wind turbine produces about 20 megawatt hours (MWh) of energy per day. On average, a household uses 16 kilowatt hours (kWh) per day
 - a One megawatt is the same as 1000 kilowatts. How many kilowatts are there in 20 megawatts?
 - b How many households can a typical wind turbine power?
 - c How many wind turbines are needed to power a town of 5000 households?

- 2 To complete this work sheet, you will need to use the trigonometric ratios sine (sin), cosine (cos) and tangent (tan). Each ratio involves two sides of a right-angled triangle:

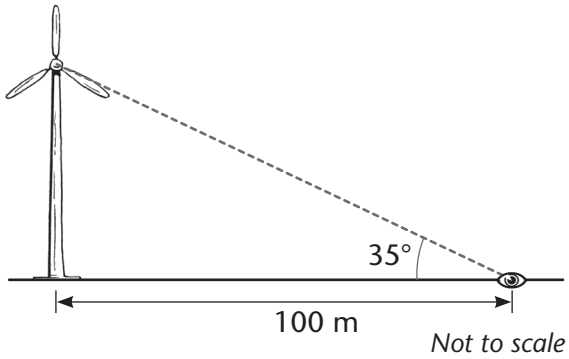


- H (hypotenuse, opposite the right angle)
- O (opposite the angle x)
- A (adjacent or next to the angle x)

SOH CAH TOA is a useful memory aid. Copy and complete the table.

SOH	CAH	TOA
$\sin x = \frac{O}{H}$	$\cos x = \frac{A}{H}$	$\tan x = \frac{O}{A}$

- 3 a Jason is 100 metres from a wind turbine. He measures the angle of elevation of the top of the tower as 35°. How tall is the tower, to the nearest metre?

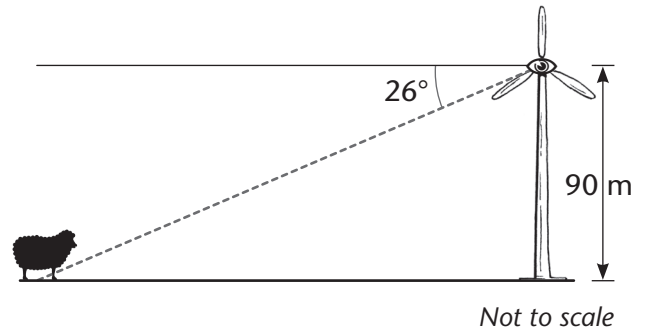


- b At the same position, Jason measures an angle of elevation of 47° to the top of a turbine blade as it reaches the top of its rotation.
 - i Calculate the height to the top of the blade to the nearest metre.
 - ii How long is the blade itself? Answer to the nearest metre.



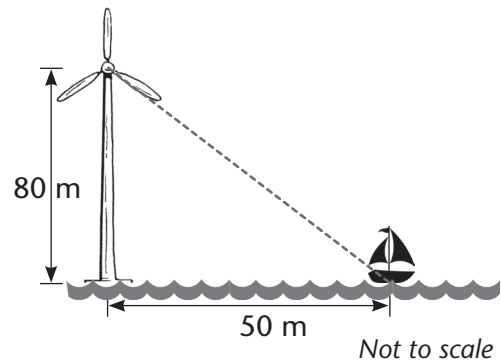
- 4** An engineer is carrying out maintenance at the top of a 90 metre wind turbine tower. She sees a distant sheep and measures an angle of depression of 26° to the sheep.

- What is the angle between the tower and a line joining the engineer and the sheep?
- How far is the sheep from the base of the tower?
Answer to the nearest metre.
- What is the straight line distance from the top of the tower to the sheep? Answer to the nearest metre.



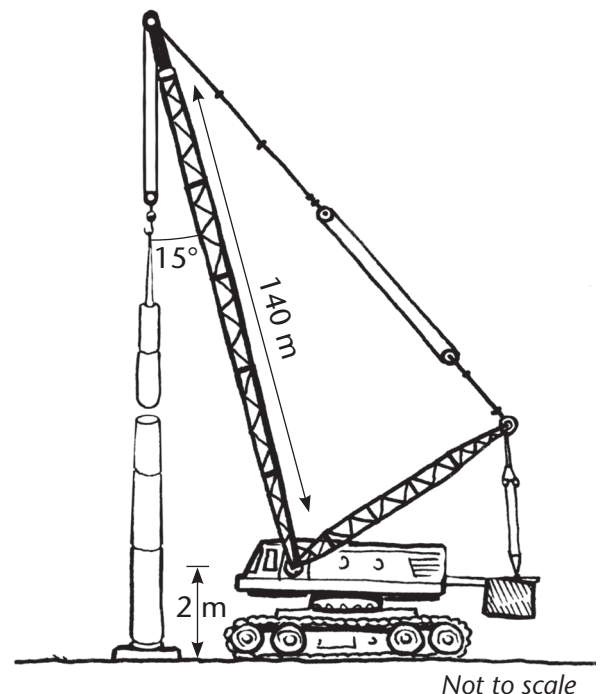
- 5** In Europe, some very large wind farms are built off-shore, in the ocean. A small boat sails near an 80 metre high wind turbine in an offshore wind farm.

- What is the angle of elevation to the top of the tower when the boat is 50 metres from the base of the tower? Answer to the nearest degree.
- How far is the boat from the tower base when the angle of elevation to the top of the tower is 72° ?
Answer to the nearest metre.



- 6** Wind turbine towers are made in sections and put together on-site using a large crane. The crane boom shown in the diagram is 140 metres long. The hoist hangs vertically at an angle of 15° to the boom.

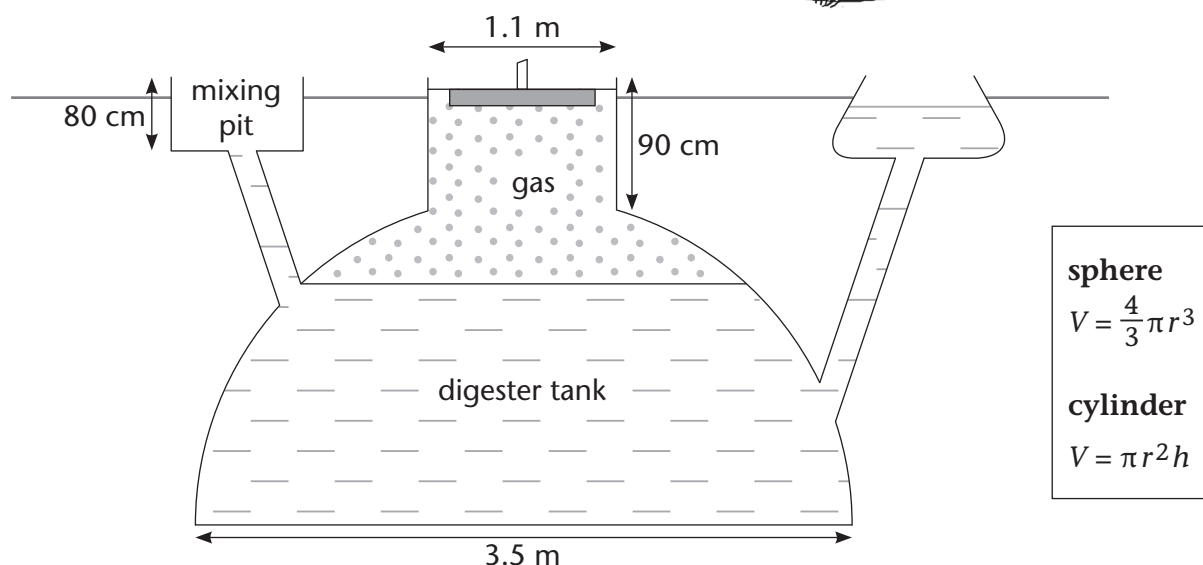
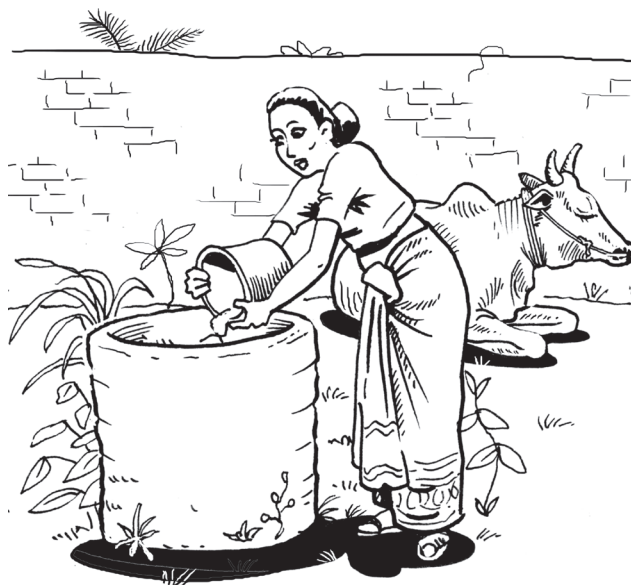
- What is the distance between the base of the boom and the centre of the wind tower?
Answer to the nearest metre.
- The crane base is 2 metres high. What is the distance from the top of the crane boom to the ground?
Answer to the nearest metre.
- The crane driver swings the crane around to pick up a tower section on a trailer that is 55 metres away from its base, and at the same height. What angle does the boom now make with the horizontal?
Answer to the nearest degree.





When waste materials such as manure, weeds and food scraps are mixed with water and stored in a tank, they decompose and produce methane gas. Methane is a biogas and a renewable source of fuel. It can be used for cooking or to power an electricity generator. Rural people in many parts of India, Africa and China use small biogas generators to provide cooking fuel. Large biogas plants are becoming more common in Europe, America and Australia.

The diagram shows a cross section of a simple biogas plant.

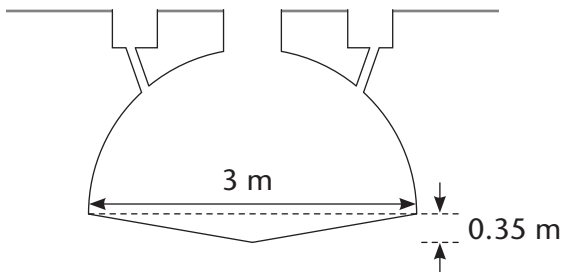


- 1 The mixing pit is a square prism (box), open at the top. The inside of the box measures 1.2 m by 1.2 m and the walls are 80 cm high .
 - a What is the wall height of the mixing pit in metres?
 - b What is the volume of the mixing pit?
 - c What is the inside surface area of
 - i the base?
 - ii one side?
 - d What is the total inside surface area of the mixing pit?

- 2 The digester tank is where the waste ferments and gas is produced. The shape of this tank is a hemisphere (half sphere) with a cylinder on top, where the gas collects. The diameter of the hemisphere is 3.5 m. The cylindrical neck has a diameter of 1.1 m and a height of 90 cm.
 - a What is the radius of the hemisphere?
 - b Calculate the volume of the hemisphere to 3 decimal places.
 - c What is the radius of the cylindrical section?



- d Calculate the volume of the cylinder to 3 decimal places.
 - e What is the total volume of the digester tank?
(Ignore the overlap of cylinder and hemisphere.)
 - f 1000 litres occupy one cubic metre.
How many litres does the tank hold, to the nearest 10 litres?
 - g When the digester contains 25% gas, how many litres is that?
- 3 In some cases, the floor of the digester tank slopes down to the centre, forming a cone. Use the measurements shown in the diagram to answer the following questions.



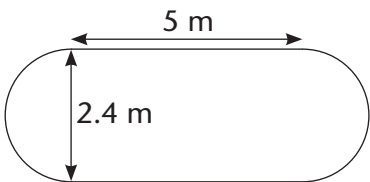
cone

$$V = \frac{1}{3} \pi r^2 h$$

- a What is the radius of the conical section at the bottom of the tank?
- b Calculate the volume of the conical section in cubic metres, to 3 decimal places.
- c What is the capacity of the conical section in litres?
- d If the whole tank has a volume of 8 500 litres, what percentage of the volume is in the conical section? Round your answer to one decimal place.

- 4 The diagram shows a different type of biogas digester tank. Formulas:

- surface area of sphere = $4 \pi r^2$
- curved surface area of cylinder = $2 \pi r L$



- a Copy and complete the table, to 2 decimal places.

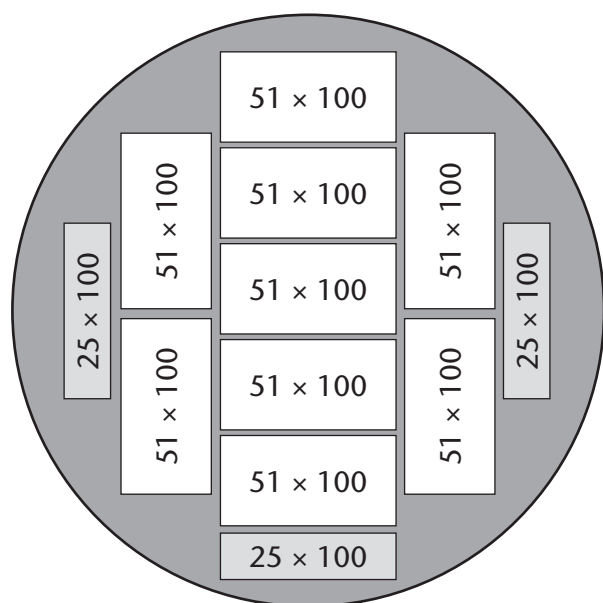
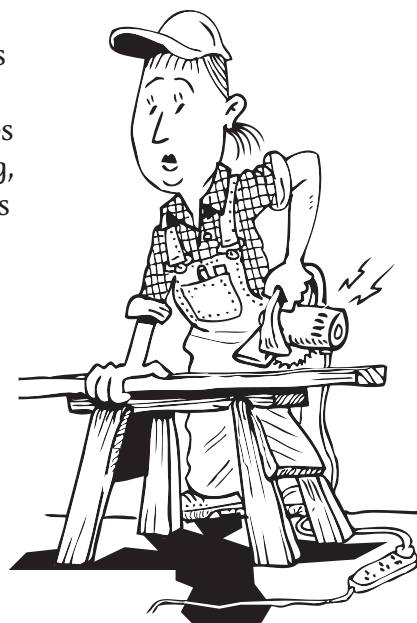
Section	Name of shape	Dimensions	Volume	Surface area
Middle		length L = diameter d = radius r =		
End		diameter d = radius r =		

- b Calculate
- i the total volume of the tank.
 - ii the total surface area of the tank.



Timber frame houses used to be built of hardwood from wild forests. These days, governments restrict the harvesting of trees from public land, to protect what is left. Most building timber comes from plantations of fast-growing pine trees. Mature trees are felled, the trunks cut into logs between 4 and 6 metres long, loaded onto a truck and taken to a mill for cutting into lengths of timber.

The diagram shows the end of a log and the cross sections of timber pieces to be cut from the log. 51×100 mm cross section timber is used as wall framing timber. The 25×100 mm can be used for decking. All measurements are in millimetres.



Source: AKD Softwoods

- 1
 - a Calculate the cross-sectional area of a 51×100 mm piece of timber.
 - b Convert this area to square metres. ($1 \text{ m}^2 = 1\,000\,000 \text{ mm}^2$)
 - c Calculate the volume of wood in one piece of timber, cross section 51×100 mm, cut from a log that is 5 metres long.
 - d Calculate the cross sectional area of a 25×100 mm piece of timber in square metres.
 - e Calculate the volume of wood in one piece of timber, cross section 25×100 mm, cut from a log that is 5 metres long.
 - f Calculate the total volume of timber that can be cut from the log that is 5 m long. Answer in cubic metres.

- 2 The log in question 1 is roughly cylindrical in shape. It has a diameter of 300 mm and is 5 m long.
 - a In square metres to 3 decimal places, what is
 - i the radius of its base?
 - ii the area of its base?
 - b What is the volume of the log in cubic metres?
 - c Use your answers to question 2b and question 1f to calculate the volume of wood that is left over after the timber has been cut from the log.



- 3** The diagram shows another cut pattern used by the mill. The log is 6 m long.

a Calculate in square metres

- the cross section area
- the volume of a 38×100 mm piece of timber.

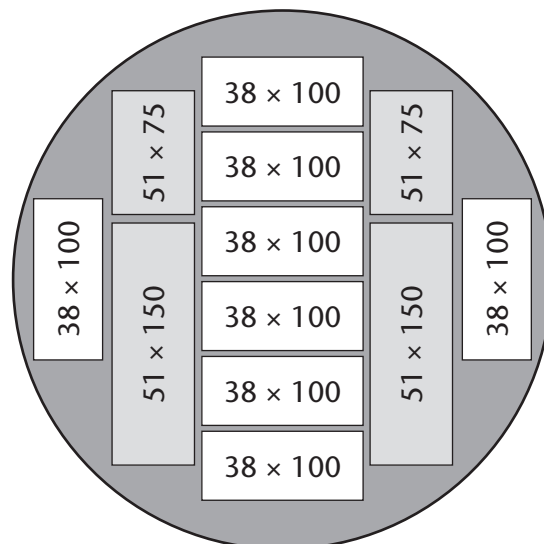
b Calculate the total volume of 38×100 mm timber cut from the 6 m log.

c The 51×100 mm pieces are used as roof beams. Calculate the volume of wood in one 51×100 mm piece.

d The diameter of the log is 320 mm. Calculate its volume in cubic metres, to 3 decimal places.

e If the total volume of usable timber in the log is 0.228 m^3 , what volume of the log is waste wood?

f What percentage of the log is waste wood?
Answer to one decimal place.



Source: AKD Softwoods

- 4** On average, 52.5% of a tree can be used for timber. Most of the waste wood is chipped and exported to Japan and China for paper manufacture. A small percentage is bark, which is stripped off the logs and sold as mulch.

A pine log has a diameter of 350 mm and is 5 m long.

a Calculate

- its volume
- its surface area.

b The bark is 5 mm thick. Calculate the volume of bark on the log.

c What percentage of the volume of the log is bark?

- 5** Once cut, the timber is dried in a kiln. Sawdust from the mill is used to fire the kilns. The dried wood is then machined to finished sizes.

For example, 51×150 mm timber is machined to 45×140 mm, which is a standard size used for roof beams.

38×100 mm is machined to 35×90 mm and used as framing timber.

Calculate to one decimal place the percentage of wood lost during the machining for

- 51×150 mm timber
- 38×100 mm timber.



Putting out the garbage bin is not everyone’s favourite chore, but we should be grateful that someone empties the bin and takes away the garbage every week. Imagine if we had to keep it all!



- 1

A typical household bin holds 120 litres of garbage.

a

If a family fills this bin every week, how many litres of garbage do they produce in a year? (One year is 52 weeks.)

b

How many cubic metres is this equivalent to? (1000 L occupies a volume of 1 cubic metre.)

c

How many years would it take to completely fill a bedroom with garbage if it is 4 metres long by 3.5 metres wide by 2.7 metres high?
Answer to the nearest year.
- 2

Fortunately, most people have regular garbage collection. Calculate how many 120 L bins of garbage will fit into a truck with a capacity of 20 m³. Round to the nearest whole number.
- 3

In densely populated countries such as England, garbage collection covers the whole country. In Australia, which has a total area of 7.692 × 10⁶ km², towns and cities occupy only 3.51 × 10⁴ km².

a

What is the remaining area of Australia?

b

What percentage of Australian land is urban (i.e. towns and cities)?
Answer correct to one decimal place.
- 4

The table shows the number of tonnes of waste delivered to landfill per year in a rural council area, for three different years.

Tonnes of waste from different sources delivered to landfill

Waste source	Year		
	2007	2010	2013
Kerbside garbage	3 390	3 622	3 772
Industrial garbage	3 061	3 260	3 474
Transfer station public garbage	1 645	1 658	1 680
Public litter bin waste	646	687	733
Municipal waste	9 926	10 145	10 812
Total waste generation	18 659	19 872	21 179

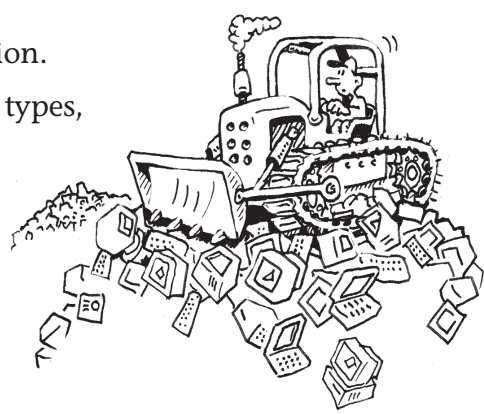


- a How many tonnes of municipal waste went to the landfill in
 - i 2010?
 - ii 2013?
- b Write your answers to part a in scientific notation. For example, 5 763 = 5.763 × 10³.
- c What is the increase in municipal waste from 2010 to 2013, in scientific notation?
- d Write in scientific notation the number of tonnes of kerbside garbage for
 - i 2007
 - ii 2010.
- e Convert your answers to part d to kilograms in scientific notation. (1 tonne = 1000 kg)
- f Write in scientific notation the amount of public litter bin waste for 2010 in
 - i tonnes
 - ii kilograms.
- g Write the tonnes of public litter bin waste for 2013 as a fraction of total waste for 2013. Convert your answer to a decimal number, round it to 3 decimal places and write it in scientific notation.
- h Write the tonnes of industrial garbage for 2013 as a fraction of total waste for 2013. Convert your answer to a decimal number, round it to 3 decimal places and write it in scientific notation.

Domestic waste received at landfills 2010					
	Organic '000 t	Non-organic '000 t	Unseparated organic and non-organic '000 t	No source type provided '000 t	Total – all types '000 t
Domestic waste	891	1 163	4 511	840	7 405
Total waste	1 702	6 534	10 575	2 815	21 626

(Based on ABS data 2009–10)

- 5 The figures in the table above are for the whole of Australia. Figures for industrial waste are not shown. All figures are in thousands of tonnes; for example, the amount of domestic organic waste written as a basic numeral is 891 000 tonnes.
- a Write the amount of domestic organic waste in scientific notation.
 - b The total amount of non-organic waste taken to landfill across Australia in 2010 was 6534 thousand tonnes. Write this
 - i as a basic numeral
 - ii in scientific notation.
 - c Write the total waste of all types
 - i as a basic numeral
 - ii in scientific notation.
 - d How many million tonnes is the total waste of all types, to the nearest million tonnes?
 - e The amount of domestic organic waste is 8.91 × 10⁵ t.
The total organic waste is 1.702 × 10⁶ t.
How many tonnes of organic waste come from non-domestic sources?
Answer in scientific notation.



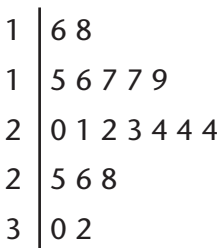


The use of recycled materials to manufacture plastic saves almost 90% of the energy that is used to manufacture plastic from its raw materials: oil and gas. Recycling just one plastic drink bottle saves enough energy to run a computer for 30 minutes.



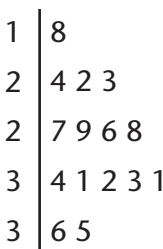
As part of their sustainable school project, the student council at Sundale High School decided to run a campaign to encourage all students to put their empty drink bottles into recycling bins.

1 Before starting their campaign, the student council ran an audit of the recycling bins around the canteen area. The stem-and-leaf plot shows the numbers of plastic bottles in the recycling bins each day for a few weeks.



- a What was the
 - i lowest number of bottles found?
 - ii highest number of bottles found?
- b The difference between the lowest and highest values is the range. Calculate the range of the numbers of plastic bottles in the recycling bins.
- c For how many days did the student council run its audit?
- d The median is the middle value of all the numbers arranged in order, from lowest to highest. What was the median number of bottles recycled over the audit period?
- e The mode is the number that occurs most often. What is the mode for this data set?

2 After the audit, the student council ran daily advertisements on the school intranet and put up posters in the canteen to encourage students to recycle plastic bottles. They then ran a second bin audit over three weeks, with the following results.



- a Rearrange the 'leaf' numbers in each row to create an ordered stem-and-leaf plot.
- b Find the median.
- c Find the mode.
- d What is the range?



3 The next term, the student council decided to continue the campaign and then run a final audit during the last three weeks of term. Here are the results of that final audit.

39, 45, 44, 43, 51, 38, 42, 34, 50, 47, 45, 52, 48, 46, 45

- a Draw a stem-and-leaf plot of these results, using stems of 3, 3, 4, 4, 5.
- b Find
 - i the median
 - ii the mode
 - iii the range.
- c Was the student council plastic bottle recycling campaign successful? Write a statement to support your answer.

4 A city council wants to choose the best locations for installing recycling bins for plastic. It hires a waste audit company to check the numbers of plastic recyclables in 30 of its large public waste bins. The results are as follows.

127, 112, 130, 127, 115, 126, 146, 122, 116, 125, 124, 105, 138, 118, 114, 150, 136, 127, 109, 116, 135, 119, 117, 141, 129, 123, 127, 131, 134, 140

- a Using stems of 10, 11, 12, etc., draw a stem-and-leaf plot of the results.
- b Find
 - i the median
 - ii the mode
 - iii the range.
- c The council decides to install recycling bins for plastic at sites where 130 or more plastic items have been counted in bins. How many sites will have plastic recycling bins installed?

5 In the recycling industry, when other materials are mixed with the recycling and have to be removed, these other materials are called 'contamination'. The daily percentage contamination volumes for two bins are shown in the back-to-back stem-and-leaf plots below.

BIN A		BIN B
	0	4 6 8 9
9 8 3	1	0 2 4 5 5 7 9
9 8 6 6 6 5 2	2	0 1 2 4
7 2 2 0	3	
0	4	
	5	



- a For bin A, what is
 - i the range?
 - ii the mode?
 - iii the median?
- b For bin B, what is
 - i the range?
 - ii the mode?
 - iii the median?
- c Which bin has less contamination?
- d Suggest a reason for the difference in the contamination levels of the two bins.



A sustainable society is one that recycles or reuses all its waste. When it comes to organic waste such as kitchen scraps and garden weeds, the best way to recycle these materials is to compost them. Anything that has been alive can be composted.

1 A healthy compost heap needs a balance between the amount of carbon (brown material) and nitrogen (green material). A good balance in a compost mixture is a carbon to nitrogen ratio of 30:1.

- a Complete the sentence:
The ratio 30:1 means that there should be ____ times as much carbon as nitrogen in a healthy compost heap.
- b Dead leaves contain about sixty times as much carbon as nitrogen. Write this as a ratio.



2 The table shows the carbon to nitrogen (C:N) ratios for some common compost materials. The best C:N ratio for compost to rot quickly is 30:1.

- a Brown materials are those that have a C:N ratio of more than 30:1. Name the brown materials listed in the table.
- b Green materials have C:N ratios of less than 30:1. List the green materials.
- c Complete the sentence:
Compost made only of paper and straw will not rot quickly because the amount of _____ is too high, compared with the amount of _____.

Carbon to Nitrogen ratios for common compost materials

Material	C:N ratio
Paper	150:1
Straw	100:1
Grass cuttings, weeds	20:1
Food waste	15:1
Cow manure	12:1

- d Lucy makes a compost pile using weeds and food waste. Which material should she add to make a good compost mix: cow manure or straw?
- e Which material has more nitrogen per kilogram: food waste or paper?

3 For many composters, the most important goal is to have roughly equal amounts of green and brown materials in their compost heap, either well-mixed or in thin layers.

- a Write the ratio of brown to green materials for the following compost heaps, in simplest form.
 - i 15 kg brown to 10 kg green
 - ii 16 kg brown to 8 kg green
 - iii 20 kg green to 18 kg brown
 - iv 2 kg brown to 2.5 kg green
 - v 2.4 kg brown to 1800 g green
- b Which of the compost heaps in part a is likely to produce the best compost?



- 4** Composting also needs water. No water means nothing rots. Too much water means slimy, smelly compost. Good compost usually needs a ratio of compost materials to water of about 2:1. How many kilograms of water should be added to these masses of materials?

- a** 4 kg
- b** 8 kg
- c** 1 kg
- d** 5 kg



- 5** These gardeners have humid conditions, where compost needs less water. The ideal ratio of materials to water is about 5:2 in a warm, damp climate.

- a** A gardener has 15 kg of materials to compost. How many kilograms of water should she add?
- b** Matilda adds 22 kg of water to her compost heap. What weight of materials should there be, for a good compost?

- 6** When compost has finished breaking down, the end product is crumbly, soil-like material which is great food for the garden. Usually, some of the material in a compost heap has to be recycled into the next heap to break down fully.

Calculate the amount of material to be recycled into a new heap for

- a** 100 kg of compost with a ratio of finished compost to material to be recycled of 4:1
- b** 45 kg of compost with a ratio of finished compost to material to be recycled of 7:2.

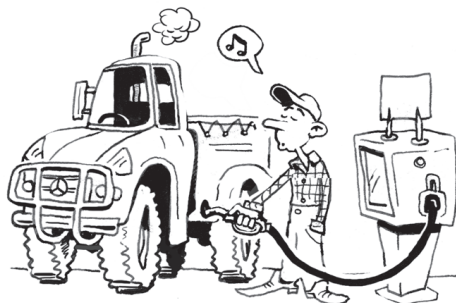
- 7** An organic farmer spreads a mixture of compost and sheep manure on a paddock with a compost to manure ratio of 5:3. If he spreads a total of 640 kg of the mixture, how much

- a** compost does he use?
- b** sheep manure does he use?

- 8** A market gardener makes 1.2 tonnes of compost.

- a** How many kilograms of compost does she have?
- b** If she decides to use compost and manure in the ratio 3:2, how many kilograms of manure will she need?

Public transport, private cars, walking and bike riding are all ways in which people in towns and cities can travel to and from school or work. The most sustainable forms of transport are ones that do not use fossil fuels such as petrol or gas. These include walking and bike riding. Public transport uses much less fossil fuel per person than a private car carrying one person.



- 1 List these forms of transport in order from least sustainable to most sustainable:
bus, bike, small car, 4WD.

- 2 Some people don't own a bike.
Not everyone has a car.
The tree diagram shows the possible combinations of car and bike ownership.

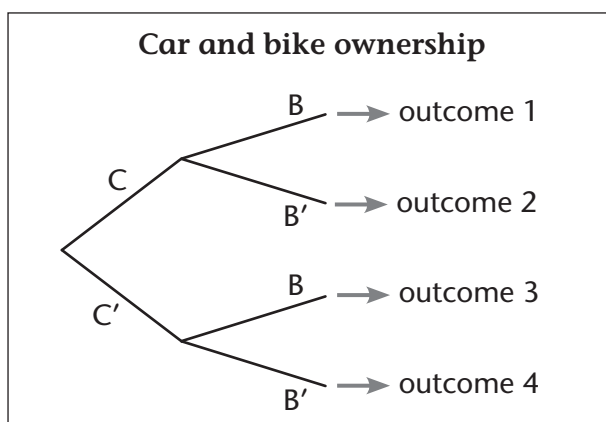
C means **owns a car**.

C' means **doesn't own a car**.

B means **owns a bike**.

B' means **doesn't own a bike**.

The top branch of the tree diagram is outcome 1, which is **owns a car and a bike**.



- a What does outcome 2 represent?
- b What does outcome 3 represent?
- c What does outcome 4 represent?

- 3 A town survey shows that 70% or 0.7 of adults own a car, and 20% or 0.2 of adults own a bike.

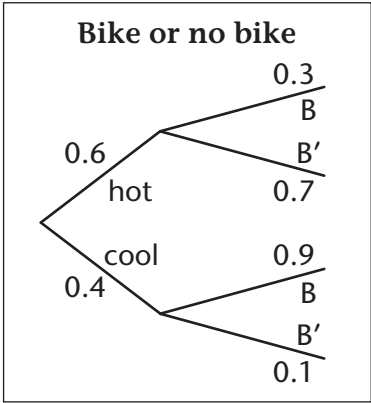
- a What is the probability, expressed as a decimal, that a person picked at random
 - i owns a car?
 - ii doesn't own a car?
- b What is the probability that a person picked at random
 - i owns a bike?
 - ii doesn't own a bike?
- c Copy the tree diagram from question 2.
Label it with the probabilities for each branch: C, C', B, B'.
- d What is the probability that an adult owns a car and a bike?
(Hint: multiply probabilities along a branch.)
- e What is the probability that an adult owns a bike but no car?





4 The weather is an important factor in how likely it is that a bike owner will choose to ride to work, instead of using a car or public transport.

The tree diagram shows the probability of hot or cool weather over summer, and the probability (B) that Grace will ride her bike to work in these different conditions.



- a What outcome does the bottom branch of the tree diagram represent?
- b What is the probability that it's a hot day and Grace rides to work?
- c What is the probability that the weather is cool and Grace rides to work?
- d What is the probability that Grace rides her bike to work on any day over summer, hot or cool?

5 Public transport is not as sustainable as bike riding, but it is much better than driving a car. In cities with good public transport systems, fewer people have cars.

A large scale survey of the public transport systems of 30 world cities revealed the results in the table at the right.

Public transport – world cities

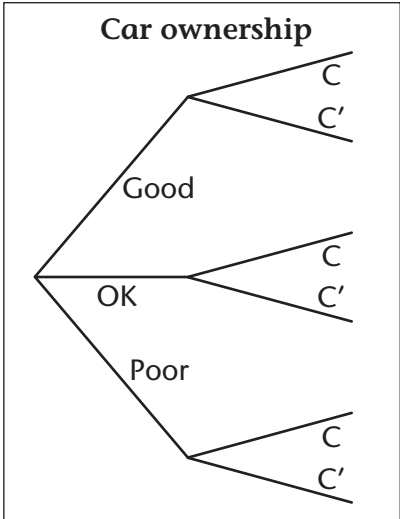
Quality of public transport	Number of cities (total 30)	Percentage of car owners
Good	3	20%
OK	15	60%
Poor	12	90%

- a Copy the table below and convert the data from the table at right to probabilities. The first row has been done for you.

Quality	P(quality)	P(car owner)
Good	$\frac{3}{30} = 0.1$	0.2
OK		
Poor		



- b Copy the tree diagram at the right and add the probabilities from your table.
- c Use the tree diagram to answer the following questions about a person picked at random from one of the cities.
 - i What is the probability that the person lives in a city with good public transport and doesn't own a car?
 - ii What is the probability that the person lives in a city with poor public transport and doesn't own a car?
 - iii What is the overall probability that the person doesn't own a car?





Travelling by air is the quickest way, and sometimes the only way, to reach a destination. Sydney to Melbourne is a one-hour air trip. Flights to holiday destinations such as Vietnam or Europe take many hours, but the same trip over land and sea takes weeks or months. The downside to such quick journeys is the large amount of greenhouse gases produced by aeroplanes. Travellers wanting to live sustainably can pay a ‘carbon offset’, which helps to fund projects that reduce carbon dioxide, such as tree planting.



- 1 Trees take carbon dioxide (CO₂) from the atmosphere and store it while they are alive.
A carbon offset company charges three dollars to plant one tree. Six trees will store, or offset, one tonne (t) of carbon.
 - a How much does it cost to offset
 - i 1 t of CO₂? ii 3 t of CO₂? iii 0.4 t of CO₂?
 - b How many tonnes of CO₂ can be offset by
 - i 60 trees? ii 300 trees?

The table shows the average amount of CO₂ produced per passenger, per kilometre, for plane flights of different distances.

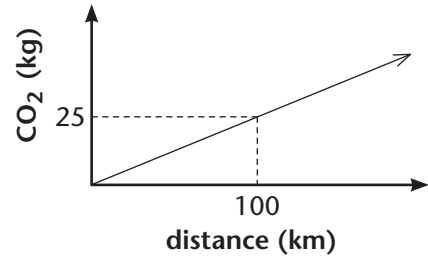
Average CO₂ produced per passenger

Flight type	kg of CO ₂ per km
Short (400 km–1000 km)	0.4
Medium (1000 km–4000 km)	0.2
Long (more than 4000 km)	0.25

- 2 Which type of flight produces
 - i the most CO₂ per kilometre?
 - ii the least CO₂ per kilometre?
- 3 Suggest reasons for the differences in the amounts of CO₂ per kilometre.
- 4 Jodi flies from Brisbane to Sydney. The flight distance is 735 km.
 - a How many kilograms of CO₂ per kilometre does she need to offset?
 - b What is the total amount of CO₂ per passenger produced by the flight, in kilograms?
 - c How many tonnes is that? (1 t = 1000 kg)
- 5 Shahid flies from Karachi to London, a flight distance of 6314 km.
 - a How many kilograms per kilometre does he need to offset?
 - b What is the total amount of CO₂ per passenger produced by the flight, in kilograms?
 - c How many tonnes is that? (1 t = 1000 kg)
 - d How much will Shahid have to pay at an offset price of \$20 per tonne?
- 6 Hannah calculates that she needs to offset 0.35 t of CO₂ produced by a medium flight.
 - a How many kilograms is 0.35 t?
 - b How many kilometres was Hannah’s flight?



- 7 The graph shows the amount of CO₂ produced per passenger against kilometres travelled, for the first part of a flight.
Is this a short, medium or long flight?



On long flights, there are different classes of seats which provide different amounts of space to sit and sleep. Economy is cheapest, and next is business class with wider seats and more leg room. Most expensive is first class with private bedrooms.

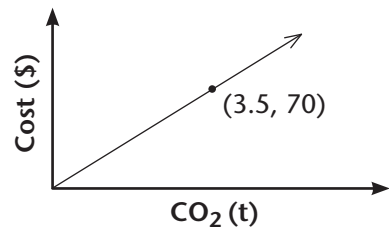
More space means a larger share of CO₂ emissions for a flight. The table shows the CO₂ emissions per kilometre for different classes on a large aircraft.

CO₂ emissions per kilometre by seating class for a large aircraft

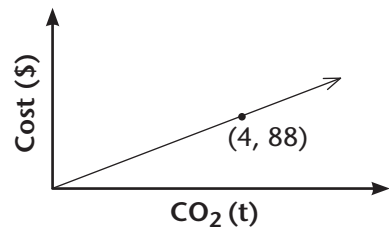
Seat class	Number of seats	CO ₂ emissions for all passengers (kg/km)	CO ₂ emissions per passenger (kg/km)
Economy	316	57	
Business	50	19	
First	14	8	

- 8 Copy the table. Calculate the CO₂ emissions per passenger for all three classes (correct to 2 decimal places) and record the values in the last column.
- 9 Correct to 2 decimal places, how many times more CO₂ emissions should a first class passenger offset than
- a a business class passenger?
 - b an economy class passenger?
- 10 a Calculate the number of tonnes of CO₂ emissions for a first class passenger on a 10 000 km flight.
- b To reach the same amount of CO₂ emissions, how many kilometres would these passengers have to fly?
- i a business class passenger
 - ii an economy class passenger
- 11 A company decides to buy carbon offsets to cover air travel by employees. The graph shows the cost against the amount of CO₂ offset.

- a The graph shows that 3.5 t of CO₂ emissions costs \$_____.
- b How much will it cost to offset 1 t of CO₂ emissions?
- c If D represents offset cost and C represents CO₂ emissions, write the equation that links D and C.



- 12 A biochar company offers carbon offsets. The graph shows the cost against the amount of CO₂ offset.
- a The graph shows that it costs \$88 to offset _____ t of CO₂ emissions.
 - b How much will it cost to offset 1 t of CO₂ emissions?
 - c If D represents offset cost and C represents CO₂ emissions, write the equation that links D and C.



A large part of the energy used by households is for heating during winter and cooling during summer. People aiming for sustainability use renewable energy to heat and cool their homes, or use as little energy as possible.

Table 1: Main source of energy for heating 2011

Proportion (%)	NSW	Vic.	Qld	SA	WA	Tas.	NT	ACT
Electricity	44.0	18.8	41.3	50.8	35.9	65.6	7.7	33.0
Gas	22.4	68.5	2.4	26.7	36.0	2.9	3.0	59.3
Wood	11.4	8.9	6.6	13.6	11.8	27.8	2.5	2.3
Other	3.2	1.6	2.2	2.1	1.4	1.8	0.6	1.6
No heater	18.9	2.2	47.5	6.8	14.9	1.9	86.3	3.9
Total	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0

(Based on ABS data 2011)

- 1
- Table 1 shows the percentage of households in each Australian state and territory that use different energy sources.

a Which is the most common source of heating used in

i NSW?

ii ACT?

b In which states is gas the most common form of heating?

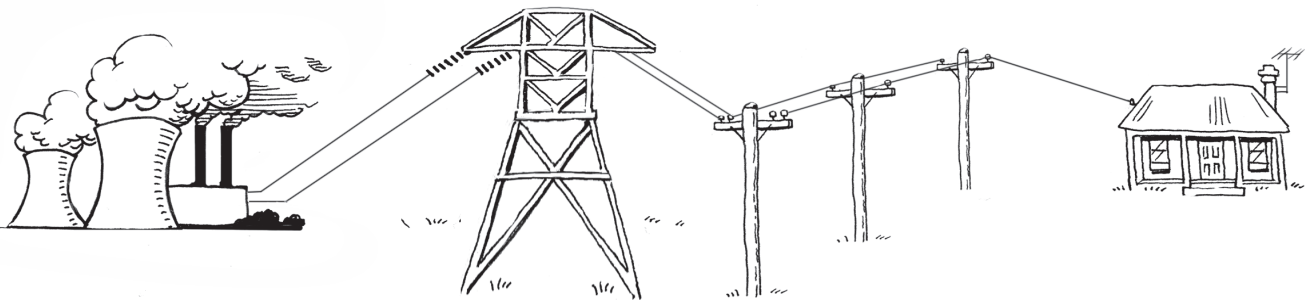
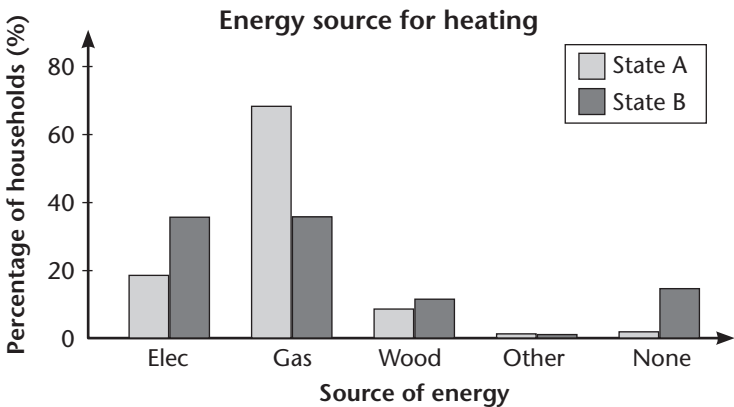
c What percentage of households use some form of heating in

i NT?

ii Tas?

d The column graph shows the percentages of different types of heating used in two states, A and B. Use the data in the table to identify state A and state B.

e Draw a column graph showing the percentages of different types of heating used in NSW.



- 2
- Use the data on electricity use across all states in Table 1 to answer the following.

a What is the mean percentage of electric heating used? Answer to one decimal place.

b What is the median percentage of electric heating used?
(Hint: average the two middle values.)



- c What are the highest and lowest percentages of electric heating used?
 - d What is the range of percentages of electric heating used?
 - e If you take out the highest and lowest percentages, what is the new
 - i mean value?
 - ii median value?
 - f What is the name given to the lowest and highest percentages in this example?
- 3 Use the Table 1 data on households with no heater across all states and territories to complete the following.
- a Round all the percentages to the nearest whole number, and arrange them from lowest to highest value.
 - b Calculate for these values
 - i the mean
 - ii the median
 - iii the mode.
 - c Which of the three results from part b is the best measure of central tendency?

Table 2: Whether cooling was used – 2005, 2011

	NSW	Vic.	Qld	SA	WA	Tas.	NT	ACT
MARCH 2011								
Dwellings with a cooler ('000)	1 759.1	1 620.5	1 265.4	606.4	773.7	92.9	76.3	96.0
Proportion of all dwellings (%)	64.2	75.5	73.6	91.3	86.3	44.4	93.9	69.5
MARCH 2005								
Dwellings with a cooler ('000)	1 391.2	1 152.1	886.2	541.0	542.5	37.7	50.3	60.0
Proportion of all dwellings (%)	53.7	60.1	57.8	84.1	68.8	19.2	91.9	47.9

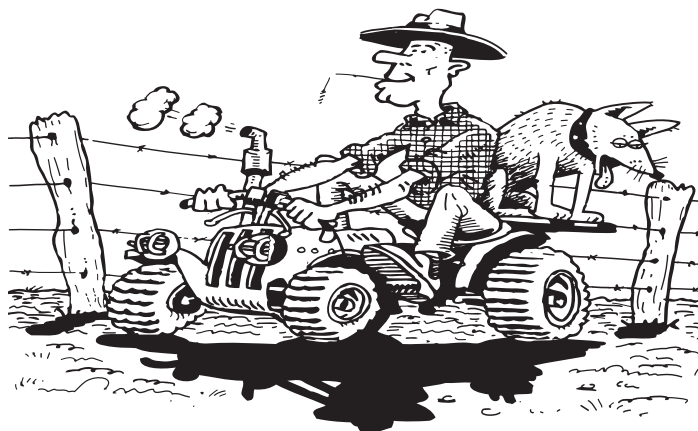
(Based on ABS data 2011)

- 4 Use the data in table 2 to answer the following questions.
- a How many dwellings in SA used cooling in
 - i 2005?
 - ii 2011?
 - b What percentage of dwellings in WA used cooling in
 - i 2005?
 - ii 2011?
 - c Of the dwellings that used cooling in 2011, which states had the
 - i highest proportion?
 - ii lowest proportion?
 - d Calculate the mean percentage of dwellings using cooling across all states for
 - i 2005
 - ii 2011.
 - e Calculate (to one decimal place) the median percentage of dwellings using cooling across all states for
 - i 2005
 - ii 2011.



Water is the most precious commodity on the planet. Sustainable water use often means using as little as possible, especially in dry countries like Australia.

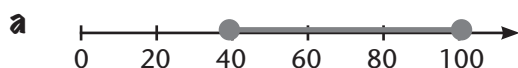
Note: in this work sheet,
 V represents volume in litres;
 t represents time in minutes.



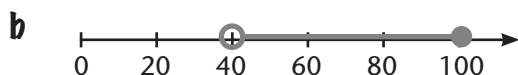
- 1 Write these inequality expressions beside the correct word statements:
 $t > 4$, $V > 500$, $V \leq 500$, $t \leq 4$.
 - a A maximum shower time of 4 minutes
 - b She wasted over 500 litres
 - c More than 4 minutes is too long
 - d They used no more than 500 litres
- 2 During a severe drought, a government enforces water restrictions. The daily maximum water use per person is 200 litres. At the same time, each person is encouraged to use at least 20 litres each day for personal health and hygiene. Which of the following inequality expressions is correct in this situation?
 - a $20 \geq V \leq 200$
 - b $20 \geq V \geq 200$
 - c $20 \leq V \leq 200$
- 3 Showering uses a lot of water in the home. The Daley family decide to reduce their water use. They agree not to use more than 75 litres of water per shower. The shower runs at a rate of 15 litres per minute.
 - a To find the maximum shower length, solve the inequality $15t \leq 75$.
 - b To further reduce their water use to no more than 60 litres per shower, the family installs a water-saver shower head which runs at 8.5 litres per minute. Write a new inequality for the shower length, using t as time in minutes.
 - c Solve the inequality from part b to the nearest minute.
- 4 Write an inequality to describe each situation.
 - a The maximum number of minutes (t) Zak can run a hose to wash his car, to avoid using more than 300 litres of water in total. The hose delivers water at 20 litres per minute. Zak uses an extra 15 litres of hot water in a bucket.
 - b The minimum number of flushes for the toilet that will use a total of more than 200 L per day. The toilet uses 12 litres of water per flush. In addition, it has a leak which uses 25 litres per day. (Use N for number of flushes.)



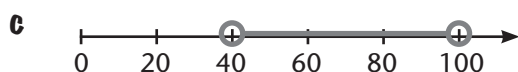
- 5** A gardener uses between 40 and 100 litres of water daily to irrigate her strawberry patch, depending on weather conditions. Select the inequality expression that applies to this situation and its corresponding number line graph.



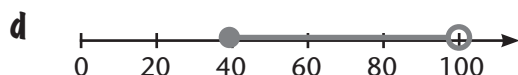
i $40 < V < 100$



ii $40 \leq V \leq 100$



iii $40 \leq V < 100$



iv $40 < V \leq 100$

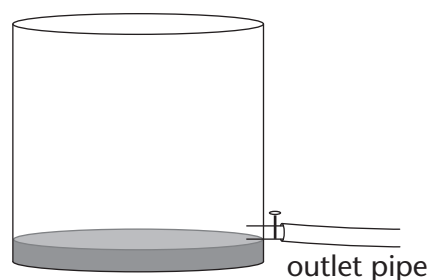


- 6** A water company charges 0.25c per litre for water used. In addition, there is a service charge of \$28 added to the bill.

- Write an expression in dollars for the cost of V litres of water. (Hint: change 0.25c to dollars.)
- Write an inequality for the maximum volume of water used that will keep the bill under \$300.
- Solve the inequality for V .
- Express the answer in kilolitres.

- 7** In rural Australia, many households rely on rainwater tanks for their house water. In times of drought, everyone must minimise their water use for the water supply to be sustainable.

One household has a 30 000 litre tank with the bottom 500 litres below the outlet pipe.

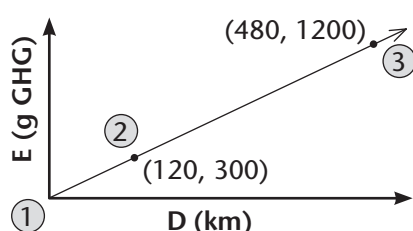


- How much water is available when the tank is full?
- Write an inequality expression that shows the volume of water possibly available to the household at a particular time.
- At the end of spring the tank is full. The rain stops suddenly and a dry summer is predicted. The family must allow for 3 months (90 days) without rain.
Write an inequality, using d as the daily water use in litres, to express the possible daily water use by the household so that there will be some water left after 90 days.
- The family decides they will also supply water to the bird bath in the garden, which takes 4 litres per day. Rewrite the inequality to include this information.
- Solve the inequality to find the maximum daily water use for the household. Answer to the nearest whole litre.



Our food often travels a long distance from where it was grown or manufactured. The term 'food miles' comes from Britain, where distance is measured in miles rather than kilometres. Transporting food uses non-renewable fuels like diesel which produce greenhouse gases. Food sources grown and made locally have low food miles, so use fewer non-renewable resources and are more sustainable.

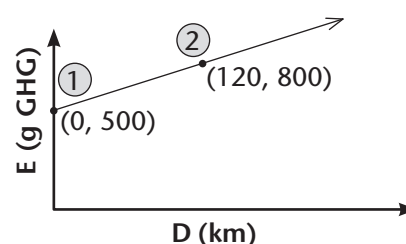
- 1 Tomatoes from north Queensland are sold all over Australia. The graph shows how the amount of greenhouse gases (GHG) emitted by a transport truck (E) increases with distance travelled (D) for each 10 kilogram box of tomatoes transported.



- a The coordinates of point 1 are (____, ____).
This point is called the o_____ of the graph.
- b Point 2 has coordinates (120, 300). This means that when $D = 120$, $E =$ _____.
When a box of tomatoes is transported 120 km, _____ grams of GHG are produced.
- c At point 3, $D =$ _____ and $E =$ _____. This means that after _____ km of travel, _____ g of GHG are produced per box of tomatoes.
- d Calculate the gradient of the line shown in the graph using the formula:
$$\text{gradient } (m) = \frac{\text{rise}}{\text{run}}$$
- e The equation of a straight line can be written as $y = mx + c$.
In this tomato transport example, the pronumeral representing the x variable is _____.
The y variable is represented by _____, the gradient $m =$ _____ and y intercept $c =$ _____.
- f Write the equation for the straight line shown in the graph.
- g Use the equation to calculate the amount of GHG produced in transporting the tomatoes to Melbourne, a distance of 3300 km.
- h After what distance will 1 kg of GHG be produced? (1 kg = 1000 g)

- 2 Growing and harvesting tomatoes also produces a fixed amount of GHG per box of tomatoes. In this graph, the fixed amount of GHG is added, showing the total GHG emissions for a box of tomatoes.

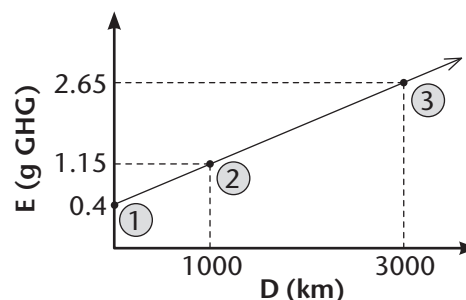
- a The coordinates of point 1 are (____, ____).
This point is called the y _____ of the graph.
- b The gradient of the graph is _____.
- c Write the equation of the graph.
- d Use the equation to calculate the amount of GHG produced after transporting the tomatoes 2700 km to Sydney.





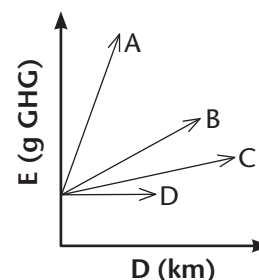
- 3** Fresh fruit is shipped between countries in refrigerated containers. The graph shows how the total GHG emissions in kilograms increases with distance transported for a 5 kg box of Californian oranges. Refrigeration is counted as part of the fixed cost.

- Point 1 is the y intercept of the graph.
Its coordinates are (____, ____)
- Point 2 has coordinates (1000, 1.15).
This means that when $D = 1000$, $E =$ _____.
When a box of oranges is transported 1000 km, _____ kg of GHG is produced.
- The coordinates of point 3 are (____, ____).
This means that when $D =$ _____, $E =$ _____.
When a box of oranges is transported _____ km, _____ kg of GHG is produced.
- Calculate the gradient of the graph.
- Write the equation of the graph.
- Calculate the total GHG emissions when the oranges are transported from Los Angeles to Adelaide, a distance of 7400 km.



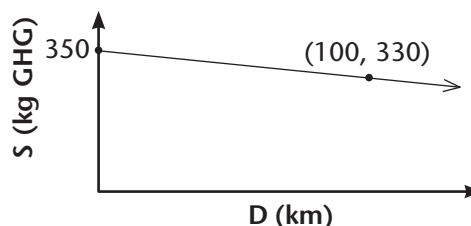
- 4** An apple is transported from one place to another. Match each line in the graph of GHG emissions against distance travelled, with the following apple transport methods.

- Travelled in a delivery truck
- Helicoptered to a mining site
- Carried by a bush walker on an expedition
- Travelled in a rail shipment



- 5** A potato farmer decides to follow organic farming methods to reduce the greenhouse gases produced by artificial fertilisers and pesticides. The graph shows how the saving (S) in kilograms of GHG produced changes with the distance (D) that the farmer's crop of potatoes is transported from the farm.

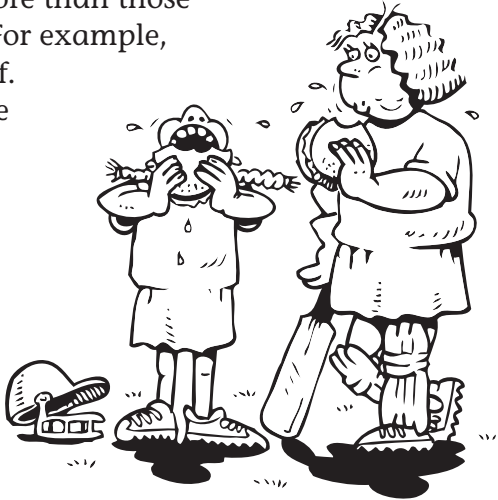
- What is the y intercept of the graph?
- Calculate the gradient of the graph.
- Write the equation of the graph.
- Use the equation to calculate how far the potatoes are transported before the saving in GHGs falls to zero.





The resources needed to produce meat are many times more than those needed to produce the same amount of vegetarian food. For example, it takes 7 kilograms of grain to produce 1 kilogram of beef. Cows burp and flatulate a lot. 40 kilograms of greenhouse gases are produced for every kilogram of beef grown. Growing animals for meat also uses over 8% of the world’s total fresh water supplies.

- 1 The table shows the amount of meat produced in 1995 and in 2007, for Australia, China and the world.
- All numbers are in thousands of tonnes (t).



Meat production
(thousands of tonnes)

Year	1995	2007
World	206 853	285 700
Australia	3 297	4 164
China	46 130	88 681

(FAO 2009)

- a How much meat did Australia produce in 2007?
- b What was the increase in world meat production between 1995 and 2007?
- c What was the increase in meat production in China between 1995 and 2007?
- d China’s percentage of world meat
- $$= \frac{\text{China meat tonnes}}{\text{World meat tonnes}} \times 100$$

What percentage of the world’s meat was produced in China in 2007?

- e What percentage of the world’s meat was produced in Australia in 2007?
- Answer to one decimal place.

- 2 The table shows the average number of kilograms of meat eaten per person in a year, for different countries.

- a In which country was the average meat consumption in 2005
- i the lowest? ii the highest?
- b Calculate the increase in the amount of meat eaten per person between 1995 and 2005 in
- i USA ii China.

Average meat consumption
(kg per person)

Country	1995	2005
Australia	105.7	117.6
UK	73.5	83.9
USA	117.1	126.6
China	38.2	59.5
Iran	22.2	30.4
India	4.7	5.1
Kenya	13	15.4

(FAO 2009)



c Percentage increase = $\frac{\text{increase}}{\text{1995 value}} \times 100$

Use your answers to part b and the formula above to find the percentage increase in meat consumption between 1995 and 2005 for

- i USA ii China.

- d Which country has the larger increase in average meat consumption: Australia or Kenya?
- e Which country has the larger percentage increase in average meat consumption: Australia or Kenya?



- 3 Much of the world's farmland is pasture for animals. The amount of pasture land worldwide has increased, as has the amount of land used for crops (arable land). Forests have decreased.

The table shows the areas of different types of land use in millions of hectares, in 1991 and 2007.

**World land use
(millions of hectares)**

	1991	2007
Pasture	3337	3378
Arable	1403	1411
Forest	4068	3937

(FAO 2009)

- a How much land was cleared for pasture between 1991 and 2007?
- b Between 1991 and 2007, what is the percentage increase in
- i pasture? ii arable land?
- c How many million hectares of forest were lost between 1991 and 2007?
- d What is the percentage decrease in forest between 1991 and 2007?
- e About 33% of arable land worldwide is used to grow crops to feed animals. How many million hectares of feed crops were grown in 2007? Answer to the nearest million.

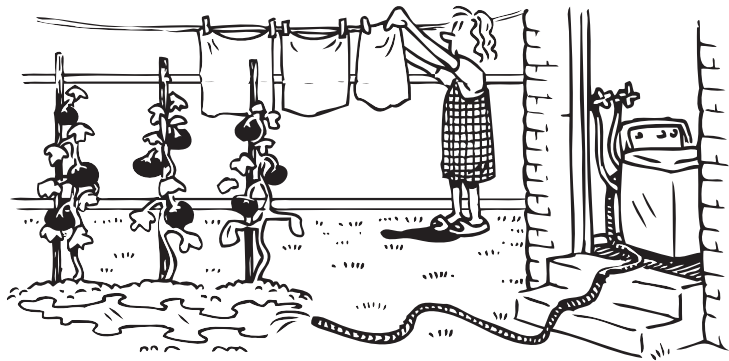
- 4 A family decides to eat vegetarian two nights per week. They calculate that this cuts the greenhouse gas (GHG) emissions of their food by 12%.

- a If their food GHG emissions were 560 kg per week, what will the new value be?
- b What would the percentage decrease in food GHG emissions be if the family ate vegetarian meals every night of the week?
- c What would the family's GHG emissions be if they ate vegetarian meals every night of the week?



Anyone who has a garden and a water supply can grow food. People without gardens can join a community garden, or share a friend's garden.

Food grown at home doesn't need transporting which reduces greenhouse gases. Home gardeners can control how much pesticide and artificial fertiliser they use, and home-grown vegetables and fruit usually taste much better than those bought in shops.



- 1 Donella wants to grow tomatoes and asks her nonna, Gina, how many seedlings she should plant. Nonna Gina says that she plants 20 seedlings each spring and harvests about 150 kg of tomatoes over summer and autumn.
- a What is the average number of kilograms of tomatoes Gina picks from one plant?

b How many kilograms can Donella expect to pick from five plants?

c Nonna Gina uses 100 kg of her tomatoes to make tomato paste. She needs 6 kg of tomatoes to make 1 kg of paste. How many kilograms of paste does Gina produce? Answer to the nearest kg.

d Donella wants to make 5 kg of paste.

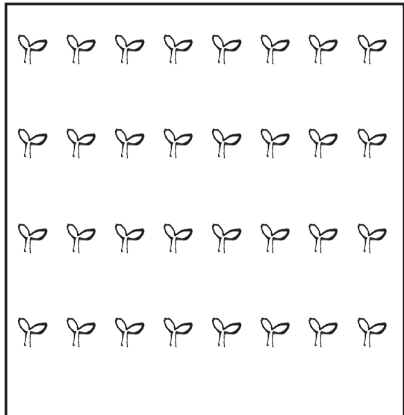
i How many kilograms of tomatoes will she need?

ii How many seedlings should she plant?

- 2 Vinh grows fresh greens in his garden to add to delicious stir-fry dinners. He plants out seedlings in rows, the same number in each row.

- a Copy and complete the table, which shows the total number of seedlings for different numbers of rows planted.

Rows	1	2	3	4	5
Seedlings		16		32	



- b How many seedlings will Vinh need for

i 8 rows?

ii 15 rows?

c Vinh grows his own seedlings by planting seeds in trays of potting mix. A 3 kg bag of potting mix is enough for 12 trays. How much potting mix does Vinh need for one tray?

d Vinh feeds his plants regularly with fertiliser. His neighbour gives him a 10 kg load of compost every two months. Vinh's garden needs 8 kg of fertiliser each month. How much extra fertiliser does he need to buy per month?

e Greens need plenty of water. Vinh estimates that he uses 3000 litres of water in a 6 week period. To the nearest litre, how much does he use

i per week?

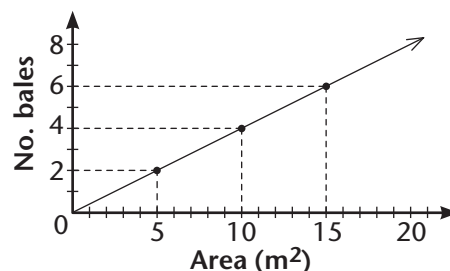
ii per day?



- 3** The Green family decides to turn their lawn into a vegetable garden. They use the 'no dig' method to prepare their garden, laying down old newspapers first to stop grass growing through, then layering straw, vegetable scraps from the greengrocer and horse manure.

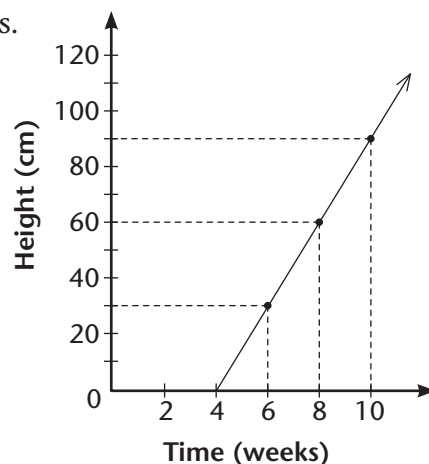
- a** The lawn is a rectangle 4 metres wide by 5 metres long.
What is its area in square metres?
- b** If it takes 6 newspapers to cover one square metre of ground, how many newspapers will they need to cover the lawn?

- c** The graph shows the number of bales of straw needed for different sizes of gardens.



- i** Use the graph to calculate the amount of straw needed per square metre of garden.
- ii** How many bales of straw does the Green family need for their garden?
- iii** The garden has a total of 3 straw layers. How many bales are needed per layer?
- d** The Greens buy 8 bags of horse manure, which is enough for two of the manure layers. How many more bags do they need for the third layer?
- e** The instructions say to use 3 times as much vegetable material as manure. How many bags of vegetable scraps will they use to make the garden?
- f** A thin layer of soil goes on last. 4 kg of soil covers 5 square metres of garden. How much soil does it take to cover the whole garden?

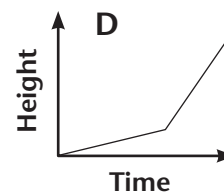
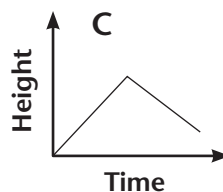
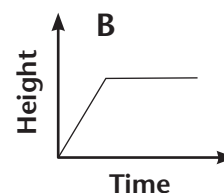
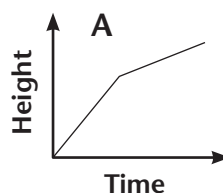
- 4** Jade loves fresh peas. She planted pea seeds next to a trellis. When the seedlings appeared, she measured the height of the pea plants as they grew. Jade drew a graph of average plant height against time in weeks.



- a** What is the average growth rate of the pea plants for the first 4 weeks?
- b** What is the average growth rate of the pea plants between 4 and 10 weeks?
- c** On average, how tall will a pea plant be at week 6?
- d** Jade grew 30 pea plants and harvested 2 kg of peas over a 5 week period.
 - i** On average, what weight of peas did Jade harvest per week?
 - ii** What average weight of peas did each plant produce over the 5 weeks?
 - iii** If Jade wants to harvest a total of 10 kg from her next crop of peas, how many pea seedlings does she need?

- 5** Match these statements to the correct graph.

- a** Donella's tomatoes grew slowly then grew faster when she added compost.
- b** Vinh's greens grew quickly for a while, then grew more slowly.
- c** Jade's pea plants grew steadily then stopped growing taller.

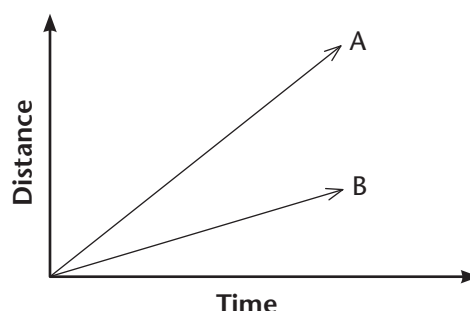




Serena has an exciting new job at a city law firm. She needs to find the best way to travel to and from work. Her options are car, bus or bike. Bike riding is her preferred, low-carbon option, but cycling is sweaty work and the showers at the office are always busy in the mornings. So Serena decides to buy herself an electric bike for commuting. The e-bike's electric motor will help her travel at a good speed, and still arrive cool and smart at the office.

- 1 The e-bike that Serena likes best costs \$2 400 or terms of 26 fortnightly repayments of \$100.
 - a How much will Serena save by paying cash?
 - b If she chooses the terms, how many payments will she need to make to pay back \$1000?
 - c Serena chooses to pay \$1000 cash and the rest on terms (26 fortnightly repayments). How much will each repayment be? Answer to the nearest cent.

- 2 Serena takes the e-bike for a test ride. The two lines on the graph, A and B, show how Serena's travel distance changes with time on her e-bike and on her old non-powered bike. Match each line to the correct bike.



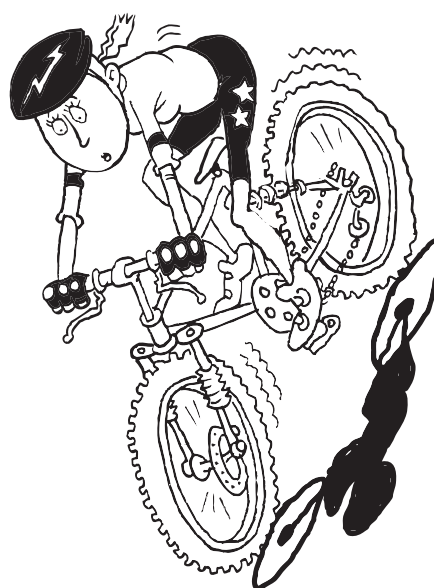
- 3 Serena travels 8 kilometres from home to the office.

- a How many minutes does it take to get to work if her average speed is
 - i 12 kilometres per hour (20 kmh^{-1})?
 - ii 20 kmh^{-1} ?
- b How fast does Serena need to travel to get to the office in
 - i half an hour?
 - ii 50 minutes?

- 4 The maximum electric power used by an e-bike is 0.25 kilowatts (kW). An electric car uses 15 kW. How many e-bikes can be operated with the amount of power needed for one car?

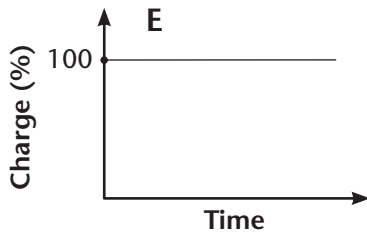
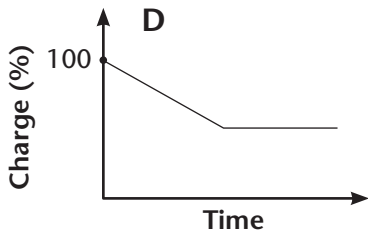
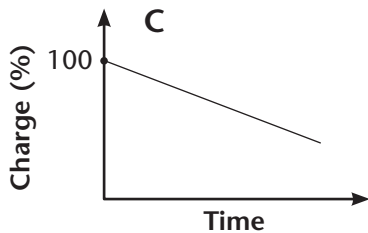
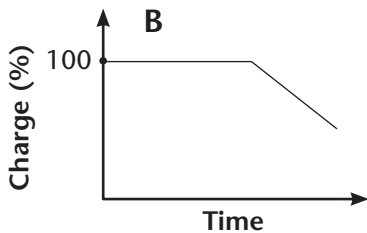
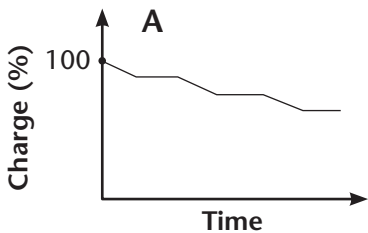
- 5 Serena takes her bike battery into the office to charge it during the day. When she first plugs the charging unit into the mains, she sees that the battery has 10% charge left. After 4 hours it is 60% charged. The battery charges at a steady rate.

- a How long will it take for the battery to be fully charged?
- b After how long will the battery be 75% charged?





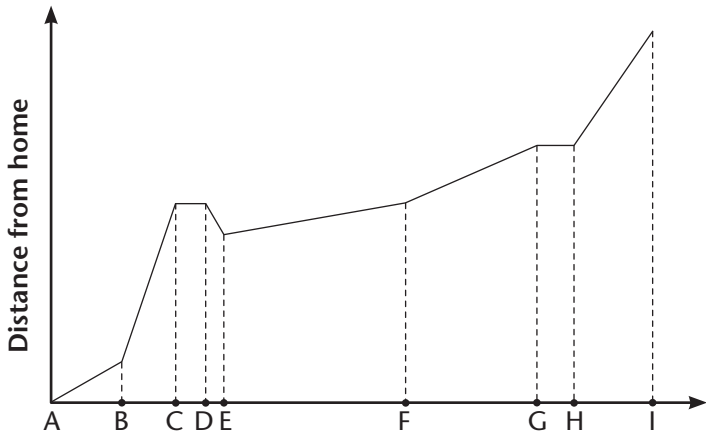
- 6 An e-bike has an electric motor, powered by a small battery that helps to turn the wheels. The motor switches on only when the rider is pedalling. The graphs show how the battery charge changes as Serena rides along a level bike path. Match each statement to the correct graph.
- a Serena pedals steadily.
 - b Serena pedals for a while then stops pedalling and coasts along.
 - c Serena does several short bursts of pedalling.



- 7 The graph shows how Serena's distance from home increases with time on her trip to work one morning.

Complete the table below to match each section of the graph with the correct statement. For example, section AB matches statement c.

- a Goes slowly uphill
- b Goes at top speed
- c Goes at medium speed
- d Waits at traffic lights
- e Turns back to read poster



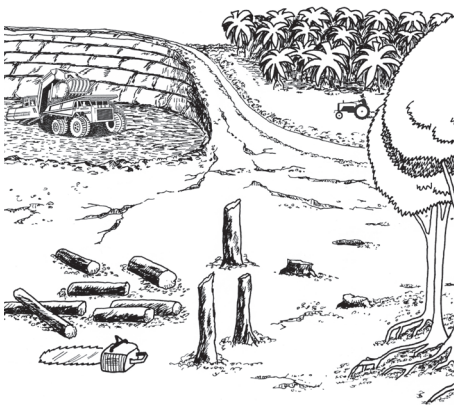
Section	AB	BC	CD	DE	EF	FG	GH	HI
Statement	c							



Forests are being cut down all over the world. Less forested area means less carbon dioxide is being taken out of the atmosphere, and less oxygen is being produced. European settlement of Australia has led to much of the forests and woodlands being cleared for farming and building. Some types of vegetation have been affected more than others. The table shows the amount of land covered by different vegetation types in 2006 and before 1750, in north Queensland.

Vegetation cover 2006 and pre-1750,
Great Barrier Reef region

Major vegetation groups	Existing Vegetation (2006) Area (ha)	Pre-1750 Vegetation Area (ha)
Acacia forests and woodlands	1 331 707	5 279 798
Callitris forests and woodlands	163 436	225 752
Casuarina forests and woodlands	54 654	69 872
Shrublands	138 775	145 612
Cleared, non-native vegetation, buildings	14 302 676	0
Eucalypt forests	3 923 813	5 801 675
Eucalypt open woodlands	3 625 166	4 296 329
Eucalypt woodlands	10 571 058	18 012 712
Heathlands	7 415	21 608
Fresh water, salt lakes, lagoons	115 265	46 287
Low, closed forests	8 881	8 908
Mangroves	219 071	226 732
Melaleuca forests and woodlands	234 261	328 292
Naturally bare – sand, rock, claypan	24 393	25 277
Other forests and woodlands	165 445	174 456
Grasslands and herblands	101 339	111 512
Rainforests	1 290 090	2 551 154
Regrowth, modified native vegetation	1 656 882	0
Sea and estuaries	8 648	25 162
Tussock grasslands	411 863	1 026 319



(Based on ABS data 2011)



- 1 Which type of vegetation that existed prior to 1750 shows the largest change in area since 1750?
- 2 Which type of vegetation shows the smallest decrease in area since 1750?
- 3 Which types of vegetation have increased in area since 1750?
- 4 Which type of vegetation covered the largest area
 - a pre-1750?
 - b in 2006?
- 5 What were the three most common vegetation types
 - a pre-1750?
 - b in 2006?
- 6 In 1750, there was no regrowth vegetation and no cleared land or buildings. Leaving these aside, what were the three least common vegetation types in 1750?
- 7 Calculate the decrease in the number of hectares between 1750 and 2006 for these vegetation types:
 - a mangroves
 - b heathlands.
- 8 There are three eucalypt vegetation groups listed in the table.
 - a Name the three groups.
 - b What was the total area of vegetation in all three eucalypt groups
 - i in 2006?
 - ii pre-1750?
 - c To one decimal place, what percentage of the total area of eucalypt vegetation was eucalypt forest
 - i in 2006?
 - ii pre-1750?
 - d To one decimal place, what percentage of the total area of eucalypt vegetation was eucalypt open woodlands
 - i in 2006?
 - ii pre-1750?
- 9 The table lists five vegetation types that are ‘forests and woodlands’. Make your own table with the headings shown below, and add the data for these five vegetation types from the table on the previous page.

Type	2006 area	1750 area	Change in area
------	-----------	-----------	----------------

- a Calculate the number of hectares lost between 1750 and 2006 for each type.
 - b Which type of forest and woodland shows
 - i the greatest change?
 - ii the least change?
 - c What is the total loss in area of forest and woodland vegetation types?
- 10 Calculate the numbers of hectares lost for
- a rainforest
 - b eucalypt forest, open woodland and woodlands combined.
(Hint: use your answers to 8b.)
- 11 Add together your results from questions 9c, 10a and 10b to find the total area of native forests and woodlands lost between 1750 and 2006.



After air, water is the natural resource that is most urgently needed by humans. Most of the water on Earth is salt water in the oceans. It is a major challenge in many countries to ensure a sustainable supply of fresh water.

Australia is a dry country. Droughts, which are already part of the climate, are predicted to become longer and more frequent as the Earth warms. Water saving is becoming an increasingly important part of living sustainably in Australia.

- 1
- The table shows the percentage of homes in Australia that have a rainwater tank.
Use the information in the table to calculate the probability (as a percentage to two decimal places) that a dwelling picked at random will have a tank, if it is:
- a

rented
- b

20 years old or more
- c

a separate house and less than 10 years old
- d

a lone-person household and rented
- e

owned outright or being purchased
- f

a townhouse that is between 10 and 20 years old
- g

a family or group household.

- 2
- In Drytown, householders have installed water saving devices. 65% of householders have installed a dual-flush toilet (DFT), 57% have installed a water-saving shower head (WSS). 32% have installed both.

Copy the Venn diagram. Use the information above to fill in the percentage represented by each area of the diagram. Then answer the following questions.

- a
- What percentage of households has a dual-flush toilet and a water-saving shower head?

b

What percentage of households has a dual-flush toilet but not a water-saving shower head?

c

What percentage of households only has a water-saving shower head?

d

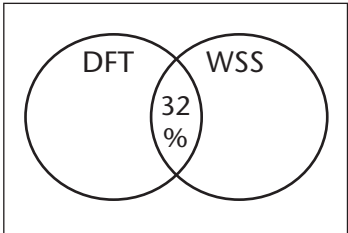
What percentage of households doesn't have either water-saving measure?

e

What is the probability that a household selected at random will have at least one water-saving measure?

Dwelling	%
Tenure	
Owned outright	35.3
Being purchased	27.8
Rented	12.6
Dwelling type	
Separate house	29.2
Terrace house, townhouse	9.6
Household type	
Family household	28.5
Lone person household	22.4
Group household	17.2
Age of dwelling	
Less than 10 years old	42.5
Between 10 & 20 years old	24.0
20 years old or more	26.7

(Based on ABS data 2010)





- 3 The two-way table shows the percentage of gardeners who use water-saving measures that cut down on evaporation: putting mulch on garden beds and watering late in the evening.

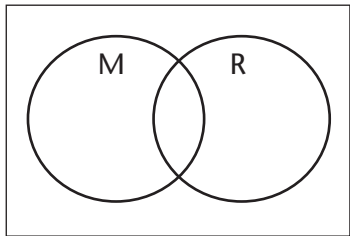
Copy the table and fill in the missing percentages.

Use the table to find the following probabilities for a gardener picked at random.

	Water late	Don't water late	Totals
Mulch	10%		32%
No mulch			
Total	26%		100%

- a Waters late and uses mulch
 - b Doesn't use mulch and doesn't water late
 - c Uses mulch but doesn't water late
 - d Uses at least one water-saving measure
- 4 The table shows the different water sources used by households in Tasmania, and the percentage of households that use each one.

Copy the Venn diagram.



Water source	%
Mains/town water	82.4
Rainwater tank	22.7
Purchased bottled water	14.6
Bore/well	3.2
River/creek/dam	5.9

(ABS 2010)

In this Venn diagram, the rectangle represents all households in Tasmania. Circle M represents the percentage that has mains water. Circle R represents the percentage that has a rainwater tank. 5.4% of households have mains water and a rainwater tank.

- a What is the percentage for the full circle M?
- b What is the percentage for the full circle R?
- c What situation is represented by the overlap of the two circles?
- d Calculate the percentages for all four areas of the diagram.
- e Use the diagram to find the following probabilities.
 - i A household with mains water does not have a rainwater tank.
 - ii A household that has a rainwater tank does not have mains water.
 - iii A household does not have mains water and does have a rainwater tank.

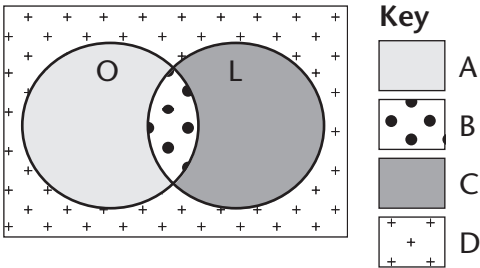
The clothes we wear are made from raw materials such as cotton, wool, hemp, nylon and PET plastic. There are energy and environmental costs to growing or making these raw materials, and manufacturing the fabric adds an extra cost. Most clothing is made in countries where wages are low, which adds a human cost. Washing clothes uses energy and water, and when we're finished with them, throwing clothes away adds an environmental cost. When all the costs are added up, our clothes cost the planet a lot more than the price we pay for them.



- 1 Over 50% of clothing is made from cotton. Growing and processing cotton uses large amounts of pesticides and bleach. Organic, unbleached cotton is grown without pesticides and is more sustainable than non-organic varieties.

The Venn diagram represents different types of T shirts sold in a store.

Set O represents T shirts made with organic cotton.
Set L represents T shirts made by workers who are paid at least enough money to live on (at least a living wage).



- a Match one or more of the four shaded areas (A, B, C or D) to each type of T shirts.
- i Made from organic cotton and made by workers who are paid at least a living wage.
 - ii Not made from organic cotton and not made by workers who are paid at least a living wage.
 - iii Made by workers who are paid at least a living wage.
 - iv Made from organic cotton.
 - v Made from organic cotton or made by workers who are paid at least a living wage, or both.
- b Which T shirts would you expect to be
- i most expensive?
 - ii least expensive?
- 2 Some methods of washing clothes use more electricity than others. The two-way table shows the percentages of people that always wash a full load, and those that wash in cold water.
- a Copy the table and calculate the missing percentages.
- b Use your completed table to find the following percentages.
- i People who always wash a full load and use hot water.
 - ii People who wash a part load and use cold water.
 - iii People who use hot water.
- c Which option is
- i least sustainable?
 - ii most sustainable?

	Hot water	Cold water	Total
Full load		25%	85%
Part load			
Total		35%	100%

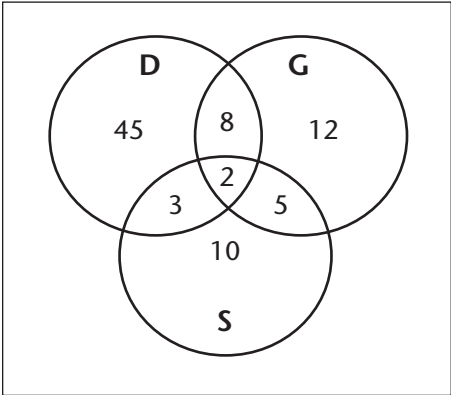


3 A sustainability group surveyed a group of 100 people on how they dispose of clothes they no longer want. The Venn diagram represents the different disposal methods.

Use the information in the diagram to find the number of people who

- a donate clothes to charity
- b sell clothes online
- c give clothes to friends and sell online, but do not donate to charity
- d donate clothes and sell online, but do not give clothes to friends
- e donate to charity, give to friends and sell online
- f do not donate to charity, give to friends or sell online.

100 people



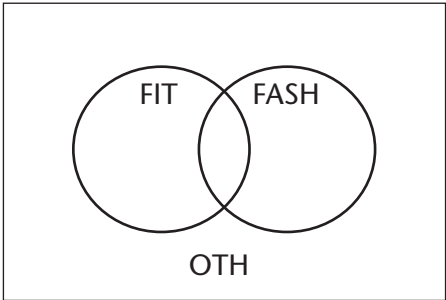
Key
Set D: donate to charity
Set G: give to friends
Set S: sell online

4 The biggest environmental savings come from making clothes last longer. Marnie organises clothes-swapping parties as part of her work for a sustainability group. People bring clothes they no longer wear and swap for 'new' clothes that other people bring.

Marnie surveys 40 people at a party to find out why they no longer want the clothes they bring. Her results are shown in the table.

Code	Reason	Number
FIT	No longer fits well	23
FASH	Not fashionable	12
OTH	Other	9

The Venn diagram can be used to represent this information.

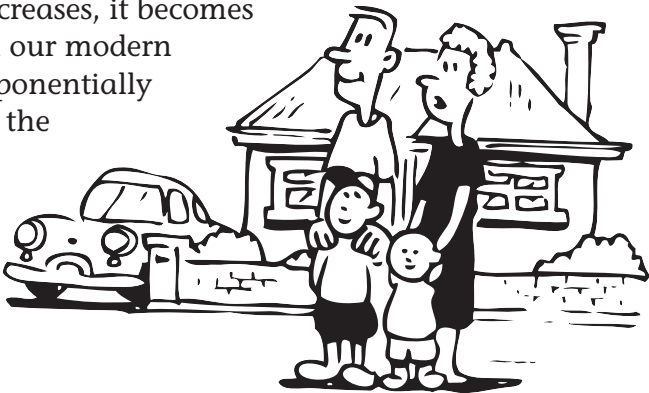


- a Copy the Venn diagram and fill in the numbers of people for each section.
- b How many people say that their clothes no longer fit and they're no longer fashionable?
- c What is the probability that a person's clothes no longer fit and they're no longer fashionable?
- d What is the probability that a person's clothes no longer fit or they're no longer fashionable, or both?





When the number of people in a community increases, it becomes harder for that community to be sustainable. In our modern world, human population has been growing exponentially for some time. In linear, or straight line growth, the same number is **added** each time period. In exponential growth, the number at the start of a time period is **multiplied** by the same number each time, to produce the number at the start of the next period. This is a much more drastic form of growth.



- 1 A simple form of exponential growth is to multiply the existing number by two, starting from 1.
 - a Use this rule to complete the sequence:
1, 2, 4, __, 16, __, 64, __
 - b We can express this rule as an exponential relationship $P = 2^t$. Copy and complete the table of values, using this rule.

<i>t</i>	0	1	2	3	4	5	6	7	8
<i>P</i>	1		4		16				

- c If *t* represents time in years, how long does it take for *P* to double in value?
 - d What is the percentage increase in *P* in one year?
- 2 Populations that grow exponentially do so at a constant rate. For example, a population has an annual growth rate of 10%. In this case, the rule we use to calculate population over time is $P = (1 + \frac{10}{100})^t$ or $P = (1.1)^t$.

- a Copy and complete the table of values to 3 decimal places, using the rule $P = (1.1)^t$.

<i>t</i>	0	1	2	3	4	5	6	7	8
<i>P</i>	1	1.1	1.21						

- b If *t* represents time in years, approximately how long does it take for *P* to double in value?
- 3 The table shows how a population *P* changes with time *t* years.

<i>t</i>	0	1	2	3	4
<i>P</i>	1	1.2	1.44	1.728	
% increase	–				

- a Copy the table and fill in the third row by calculating the percentage increase in *P* each year. For example, the percentage increase from year zero to year one is $\frac{1.2 - 1}{1} \times 100$.
 - b Complete the rule connecting *P* and *t*: $P = (\text{---})^t$



- c What is the value of P when $t = 4$?
- d How long does it take for the population to double?

4 An example of a non-exponential relationship is $P = t^2$.

a Copy the table below and use the rule $P = t^2$ to fill in the values of P .

t	1	2	3	4	5	6	7	8	9	10
P	1	4		16						

b Use the values in the table to estimate how long it will take for P to double, from these starting times.

- i $t = 2$
- ii $t = 5$
- iii $t = 7$

c Comment on doubling time in this example, compared with questions 2 and 3.

5 A quick way to find how long it will take for a population to double in size is to use the 'rule of 70':

Doubling time $\approx \frac{70}{k}$

where k is the percentage growth rate.

For example, in 2011 Thailand's annual growth rate was 0.6%.

Doubling time $= \frac{70}{0.6}$
 $= 116.67$ years

or by the year 2127.

Use the 'rule of 70' to complete the table at the right.

6 We can use the rule of 70 to calculate a growth rate (k), if we know the doubling time (D).

a Rearrange the formula $D = \frac{70}{k}$ to make k the subject.

b The human population of the world reached 7 billion people in 2011. The population was 3.5 billion, or half as many people, in 1966. What was the doubling time?

c What was the approximate annual growth rate of the world's population between 1966 and 2011? Answer correct to 2 decimal places.

d If the world's population growth rate stays steady, when will it reach

- i 14 billion?
- ii 28 billion?

e What do you think is the biggest human population that Earth can sustain? Research your answer using resources on the internet.

Population doubling times by country

Country	2011 Annual growth rate (%)	Doubling time (to 2 decimal places)	Year in which population will be double 2011 level
Australia	1.2		
Brazil	0.9		
Canada	1.0		
China	0.5		
Germany	0.0	–	never
Japan	0.3		
Timor-Leste	2.9		
United Kingdom	0.8		
United States	0.7		

(The World Bank: Population growth (annual %))



Mitch and Celeste are the owners of 'Precious Earth' organic café. They decide to stop selling bottled water from their drinks fridge. Celeste has discovered that it takes seven litres of water and 1.5 megajoules of energy to produce a plastic bottle.

- 1 To fill the space in the drinks fridge, Mitch chooses two popular drinks that come in glass containers: Aurora (\$5) and Bright (\$6). He calculates that there will be space each week for an extra 1100 drinks, and that they need to make \$6000 per week from sales to cover the lost sales of water. How many of each drink do they need to sell?

a First equation:

Use a as the number of Aurora drinks,
 b as the number of _____ drinks.

$$a + b = 1100 \quad (1)$$

b Second equation:

One Aurora costs \$5, a Auroras cost $a \times \$5$ or $5a$

One Bright costs \$6, b Brights cost $b \times \$6$ or _____

$$5a + 6b = 6000 \quad (2)$$

- c (1) and (2) are called sim__aneous equations.

d Solve using a graph

Find the intercepts of each line.

$$a + b = 1100 \quad (1)$$

When $a = 0$: $0 + b = 1100$, i.e. $b = \underline{\hspace{2cm}}$

When $b = 0$: $a + 0 = 1100$, i.e. $a = \underline{\hspace{2cm}}$

$$5a + 6b = 6000 \quad (2)$$

When $a = 0$: $5 \times 0 + 6b = 6000$, i.e. $6b = \underline{\hspace{2cm}}$ so $b = \underline{\hspace{2cm}}$

When $b = 0$: $5a + 6 \times 0 = 6000$, i.e. $5a = \underline{\hspace{2cm}}$ so $a = \underline{\hspace{2cm}}$

Lines (1) and (2) on the graph have been drawn by joining their intercepts. From the graph, estimate the coordinates of the point where the two lines cross.

$$a = \underline{\hspace{2cm}} \quad b = \underline{\hspace{2cm}}$$

So the café needs to sell _____ Aurora
 and _____ Bright drinks

e Solve by elimination

$$a + b = 1100 \quad (1)$$

$$5a + 6b = 6000 \quad (2)$$

To eliminate a , multiply equation (1) by 5.

$$5a + \underline{\hspace{1cm}}b = \underline{\hspace{2cm}} \quad (3)$$

Take equation (3) from equation (2).

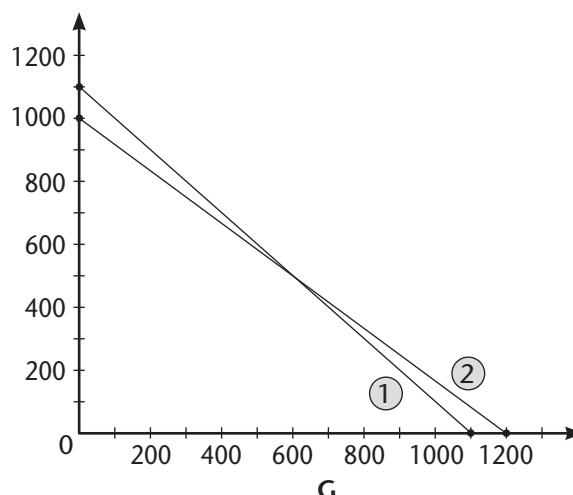
$$b = \underline{\hspace{2cm}}$$

Substitute value for b in equation (1).

$$a + \underline{\hspace{1cm}} = 1100$$

therefore $a = \underline{\hspace{2cm}}$

So the café needs to sell _____ Aurora and _____ Bright drinks.





- 2** Celeste realises the café will need more water jugs and glasses. She calls a supplier who offers 10 jugs and 100 glasses for \$250, or 15 jugs and 80 glasses for \$235. What are the prices of one jug and one glass?

- a** Use the pronumerals J and G to represent prices of a jug (J) and a glass (G). Write two equations that express the information given.

$$\underline{\hspace{2cm}}J + 100G = 250 \quad (1)$$

$$15J + \underline{\hspace{2cm}}G = \underline{\hspace{2cm}} \quad (2)$$

- b** Solve using a graph

Find intercepts of line 1.

When $J = 0$, $100G = 250$, $G = \underline{\hspace{2cm}}$

When $G = 0$, $10J = 250$, $J = \underline{\hspace{2cm}}$

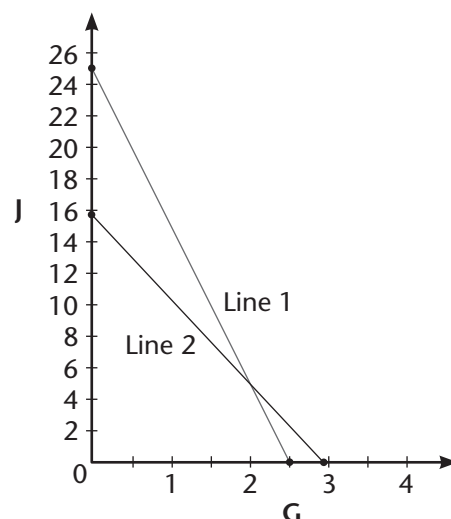
Find intercepts of line 2.

When $J = 0$, $80G = 235$, $G = \underline{\hspace{2cm}}$

When $G = 0$, $15J = 235$, $J = \underline{\hspace{2cm}}$

Lines (1) and (2) on the graph have been drawn by joining their intercepts. From the graph, estimate the coordinates of the point where the two lines cross.

Lines cross at $J = \underline{\hspace{2cm}}$, $G = \underline{\hspace{2cm}}$
i.e. a jug costs \$ $\underline{\hspace{2cm}}$ and a glass costs $\underline{\hspace{2cm}}$.



- c** Solve by elimination

$$10J + 100G = 250 \quad (1)$$

$$15J + 80G = 235 \quad (2)$$

To eliminate J , multiply equation (1) by 3 and equation (2) by 2.

$$30J + 300G = \underline{\hspace{2cm}} \quad (3)$$

$$30J + 160G = \underline{\hspace{2cm}} \quad (4)$$

Take equation (4) from equation (3).

$$\underline{\hspace{2cm}}G = \underline{\hspace{2cm}}$$

$$G = \underline{\hspace{2cm}}$$

To find J , substitute value for G in (1) or (2) and solve for J .

Solution: $G = \underline{\hspace{2cm}}$, $J = \underline{\hspace{2cm}}$

i.e. a jug costs \$ $\underline{\hspace{2cm}}$ and a glass costs $\underline{\hspace{2cm}}$.

- 3** Mitch and Celeste decide to order some publicity material to promote their bottled-water-free zone. They choose staff aprons with a 'Say No to The Bottle' logo and give away postcards that come in packs of 100.

The postcards explain why using tap water cuts down on environmental costs.

20 aprons and 50 packs of postcards cost \$1300.

10 aprons and 100 packs of postcards cost \$1400.

How much is one apron and one pack of postcards?

- a** Write a pair of equations that express the information.
b Use a graphing technique to find an approximate solution.
c Solve the equations by elimination.



The planet we live on has taken a very long time to become a place where humans can live. We take for granted the fact that life forms such as air-breathing, complex organisms like humans can exist on Earth. It has not always been so.

For most of the life of our planet, conditions have not been suitable for any life forms, except in the oceans.



By looking at the immense time scales involved in the evolution of life on Earth, we can see how important it is to live sustainably on the planet to keep conditions right for human existence. It's a long way to the next human-habitable planet.

- 1** This work sheet is about working with large numbers written in scientific notation. A number written in scientific notation has two parts: a number that is one or more, but less than ten, multiplied by a power of ten.

For example, we can write ninety six million as a basic numeral 96 000 000, or in scientific notation as 9.6×10^7 . Four thousand can be written as 4000 (basic numeral) or 4×10^3 (scientific notation).

- a** Write forty two million as
 - i** a basic numeral
 - ii** a number in scientific notation.
- b** One billion is one thousand million. Write one billion as
 - i** a basic numeral
 - ii** a number in scientific notation.
- c** Which of these numbers are correctly written in scientific notation?
 - i** 2.5×10^4
 - ii** 13.6×10^6
 - iii** 8×10^5
 - iv** 0.753×10^8

- 2** Scientists estimate Earth to be four billion, six hundred million years old.

- a** Write this number as a basic numeral.
- b** Fill in the gaps to write the age of Earth as a number in scientific notation:
 $4._ \times 10__$
- c** Complete the sentence: *Earth is four point ____ billion years old.*
- d** Scientists believe that a collision between Earth and a second planet named Theia led to the moon being formed. This happened four point five billion, or $4._ \times 10__$ years ago.
- e** How long after Earth formed did the collision happen? Write the answer
 - i** in words
 - ii** as a basic numeral
 - iii** as a number in scientific notation.



- 3** Life began to evolve on Earth during the Proterozoic era, which lasted from 2420 million years ago to 541 million years ago.
- Write these two numbers in scientific notation.
 - How long did the Proterozoic era last? Write your answer
 - as a basic numeral
 - in scientific notation.
 - Single-cell life forms have been found in rocks that are 1800 million years old. How long after the start of the Proterozoic era is this? Answer in scientific notation.
 - The first multi-cell life forms appeared one billion years ago. How many years is this before the end of the Proterozoic era? Answer in scientific notation.
- 4** An event called 'The Great Dying' occurred at the end of the Permian period (252 million years ago). Almost all life on Earth became extinct due to a large rise in CO_2 in the atmosphere. Massive eruptions of lava in Siberia are thought to have been responsible.
- The eruptions produced 1.5 million cubic kilometres of lava. Write this in scientific notation.
 - One kilometre = 1000 metres. Volume = length $\times$ width $\times$ height. How many cubic metres are there in one cubic kilometre? Answer in scientific notation.
 - Write the volume of lava from part a in cubic metres, in scientific notation.
- 5** Butterflies first appeared on Earth 40 million years ago. Modern humans evolved 2 hundred thousand years ago. How many years before humans did the first butterflies appear? Write your answer
- as a basic numeral
 - in scientific notation.
- 6** There are probably many human-habitable planets like Earth in our galaxy, but a long way off. One possibility is a planet that orbits the star Gliese 667C. It might have air and water. It is 22 light years from Earth. One light year is the distance travelled by light in one year.
- Calculate the number of seconds in one year.
(3600 seconds = 1 hour, 24 hours = 1 day, 365 days = 1 year).
Answer in scientific notation.
 - Light travels at 3×10^5 km per second.
Use your answer from part a to find how many kilometres there are in a light year.
Answer in scientific notation.
 - Write the distance from Earth to Gliese 667C in kilometres, in scientific notation.
 - Mars is the next planet from the Sun after Earth. When Earth and Mars are closest to each other, the distance between them is about 54.6 million kilometres. Which option correctly completes the statement?
The distance from Earth to Gliese 667C is about _____ times as far as Mars.
 - 38 thousand
 - 38 million
 - 38 billion
 - If it takes 6 months to get to Mars, how long would it take to travel to Gliese 677C at the same average speed?
 - Name at least two important differences between Mars and Earth.

Selected answers



For complete answers including diagrams, refer to the PDF document on the CD-ROM.

1 Solar power

- 1 a 10 b 30
2 2 kW
3 a 1 653 300 mm² b 1.67 m by 0.9 m c 1.653 3 m²
d 1 000 000 (one million)
4 a 1.98 m² b 19.8 m² c 240 m² d 8.25% e 27
f 108
5 a 18 b 3 c 54 d 1.264 m² e 68.256 m²
f 6.744 m²
6 a 16 m² b 6.48 m² c 40.5% d 2.7 m e 5.4 m
f 7.29 m²

2 Where the wind blows

- 1 a 20 000 b 1250 c 4
2 $\sin x = \frac{O}{H}$, $\cos x = \frac{A}{H}$, $\tan x = \frac{O}{A}$
3 70 m b i 107 m ii 37 m
4 a 64° b 185 m c 206 m
5 a 58° b 26 m
6 a 36 m b 137 m c 67°

3 Biogas power

- 1 a 0.8 m b 1.152 m³ c i 1.44 m² ii 0.96 m²
d 5.28 m²
2 a 1.75 m b 11.225 m³ c 0.55 m d 0.855 m³
e 12.08 m³ f 12 080 L g 3 020 L
3 a 1.5 m b 0.825 m³ c 825 L d 9.7%
4

Middle	cylinder	$L = 5$ m $d = 2.4$ m $r = 1.2$ m	22.62 m ³	37.70 m ²
End	hemisphere	$d = 2.4$ m $r = 1.2$ m	3.62 m ³	9.05 m ²

b i 29.86 m³ ii 55.8 m²

4 Forests forever

- 1 a 5 100 mm² b 0.005 1 m² c 0.025 5 m³
d 0.002 5 m² e 0.012 5 m³ f 0.267 m³
2 a i 0.15 m ii 0.071 m² b 0.355 m³ c 0.088 m³
3 a i 0.003 8 m² ii 0.022 8 m³ b 0.182 4 m³
c 0.030 6 m³ d 0.483 m³ e 0.255 m³ f 52.8%
4 a i 0.481 m³ ii 5.498 m² b 0.027 5 m³ c 5.7%
5 a 17.6% b 17.1%

5 A load of rubbish

- 1 a 6240 litres b 6.24 m³ c 6 years
2 167 bins
3 a 7.656 9 × 10⁶ km² b 0.5%
4 a i 10 145 t ii 10 812 t
b i 1.014 5 × 10⁴ t ii 1.081 2 × 10⁴ t
c 6.67 × 10² t
d i 3.39 × 10³ t ii 3.622 × 10³ t
e i 3.39 × 10⁶ kg ii 3.622 × 10⁶ kg
f i 6.87 × 10² t ii 6.87 × 10⁵ kg g 3.5 × 10⁻²
h 1.64 × 10⁻¹
5 a 8.91 × 10⁵ b i 6 534 000 ii 6.534 × 10⁶
c i 21 626 000 ii 2.162 6 × 10⁷ d 22 e 8.11 × 10⁵

6 Recycling plastic

- 1 a i 16 ii 32 b 16 c 19 d 22 e 24
2 b 29 c 31 d 18
3 b i 45 ii 45 iii 18
c Yes, the median has increased from 22 to 45 and mode from 22 to 45. The average number of bottles recycled has more than doubled.
4 b i 126.5 ii 127 iii 45 c 10
5 a i 27 ii 26 iii 26 b i 20 ii 15 iii 15 c Bin B
d Discuss with teacher.

7 Let it rot

- 1 a 30 b 60:1
2 a paper, straw b grass cuttings, weeds, food waste, cow manure c carbon, nitrogen d straw e food waste
3 a i 3:2 ii 2:1 iii 9:10 iv 4:5 v 4:3 b iii
4 a 2 kg b 4 kg c 0.5 kg d 2.5 kg
5 a 6 kg b 55 kg
6 a 20 kg b 10 kg
7 a 400 kg b 240 kg
8 a 1200 kg b 800 kg

8 On yer bike

- 1 4WD, small car, bus, bike
2 a owns car but no bike b doesn't own car, owns bike
c doesn't own car or bike
3 a i 0.7 ii 0.3 b i 0.2 ii 0.8 d 0.14 e 0.06
4 a cool weather, Grace doesn't ride to work b 0.18
c 0.36 d 0.54
5

Quality	P(quality)	P(car owner)
Good	0.1	0.2
OK	0.5	0.6
Poor	0.4	0.9

c i 0.08 ii 0.04 iii 0.32

9 Flying carbon neutral

- 1 a i \$18 ii \$54 iii \$7.20 b i 10 t ii 50 t
2 a i short ii medium
3 Discuss with teacher.
4 a 0.4 b 294 kg c 0.294 t
5 a 0.25 b 1578.5 kg c 1.578 5 t d \$31.57
6 a 350 kg b 1750 km
7 long
8

Seat class	CO ₂ per passenger
Economy	0.18
Business	0.38
First	0.57

- 9 a 1.5 b 3.17
10 a 5.7t b i 15 000 km ii 31 667 km
11 a \$70 b \$20 c D = 20C
12 a 4 t b \$22 c D = 22C

Selected answers (cont'd)



10 Comfort zone

- 1 a i Electricity ii Gas b Vic, WA, ACT
c i 13.7 ii 98.1 d A: Vic., B: WA
- 2 a 37.1 b 38.6 c 65.6, 7.7 d 57.9
e i 37.3 ii 38.6 f outliers
- 3 a 2, 2, 4, 7, 15, 19, 48, 86 b i 22.9% ii 11 iii 2
c median
- 4 a i 541,000 ii 606,400 b i 68.8 ii 86.3
c i NT ii Tas d i 60.44 ii 74.8 e i 59.0 ii 74.6

11 Don't let the tap run dry

- 1 a $t \leq 4$ b $V > 500$ c $t > 4$ d $V \leq 500$
- 2 c
- 3 a $t \leq 5$ b $8.5t \leq 60$ c $t \leq 7$
- 4 a $20t + 15 \leq 300$ b $12N + 25 > 200$
- 5 c, i
- 6 a 0.002 5V b $0.002\ 5V + 28 < 300$ c 108 800 L
d 108.8 kL
- 7 a 29 500 L b $V \leq 29\ 500$ c $90d < 29\ 500$
d $90d + 360 < 29\ 500$ e $d < 324\ L$

12 Food miles

- 1 a (0, 0), origin b $E = 300, 300\ g$
c $D = 480, E = 1200, 480\ km, 1200\ g$ d 2.5
e D, E, 2.5, 0 f $E = 2.5D$ g 8 250 g h 400 km
- 2 a (0, 500), intercept b 2.5 c $E = 2.5D + 500$
d 7 250 g
- 3 a (0, 0.4) b $E = 1.15, 1.15\ kg$ c (3000, 2.65),
 $D = 3000, E = 2.65, 3000, 2.65$ d 0.000 75
e $E = 0.000\ 75D + 0.4$ f 5.95 kg
- 4 a Line B b Line A c Line D d Line C
- 5 a 350 b -0.2 c $S = -0.2D + 350$ d 1 750 km

13 The price of meat

- 1 a 4 164 000 t b 78 847 000 t c 42 551 000 d 31%
e 1.46%
- 2 a i India ii USA b i 9.5 kg ii 21.3 kg
c i 8.1% ii 55.8% d Australia e Kenya
- 3 a 41 000 000 ha b i 1.2% ii 0.6%
c 131 million ha d 3.2% e 1233 million ha
- 4 a 492.8 kg b 42% c 235.2 kg

14 Veggie gardens

- 1 a 7.5 kg b 37.5 kg c 17 kg d i 30 kg ii 4
- 2 a

Rows	1	2	3	4	5
Seedlings	8	16	24	32	40

b i 64 ii 120 c 0.25 kg d 3 kg e i 500 L ii 71 L

- 3 a $20\ m^2$ b 120 c i 0.4 bale ii 8 bales
iii 2.67 or $2\frac{2}{3}$ bales d 4 bags e 36 f 16 kg
- 4 a 0 b 15 cm/week c 30 cm
d i 0.4 kg ii 0.067 kg iii 150
- 5 a Graph D b Graph A c Graph B

15 Electric pedal power

- 1 a \$200 b 10 c \$53.85
- 2 electric – Line A, non-powered – Line B
- 3 a i 40 min ii 24 min b i $16\ kmh^{-1}$ ii $9.6\ kmh^{-1}$
- 4 60
- 5 a 3 h 12 min b 1 h 12 min

- 6 a Graph C b Graph D c Graph A

7

Section	AB	BC	CD	DE	EF	FG	GH	HI
Statement	c	b	d	e	a	c	d	b

16 We all need trees

- 1 Eucalypt woodlands
- 2 Low, closed forests
- 3 Cleared, non-native, buildings; freshwater, salt lakes, lagoons; regrowth, modified native vegetation
- 4 a Eucalypt woodlands
b Cleared, non-native, buildings
- 5 a Eucalypt woodlands, eucalypt forests, acacia forests and woodlands
b Cleared, non-native, buildings, eucalypt woodlands, eucalypt forest
- 6 Low, closed forests, heathlands, sea and estuaries
- 7 a 7 661 ha b 14 193 ha
- 8 a forests, open woodlands, woodlands
b i 18 120 037 ii 28 110 716
c i 21.7% ii 20.6% d i 58.3% ii 15.3%
- 9 b i Acacia ii Other c 4 128 667 ha
- 10 a 1 261 064 ha b 9 990 679 ha
- 11 15 380 410 ha

17 Precious water

- 1 a 12.6% b 26.7% c 12.41% d 2.82% e 63.1%
f 2.30% g 45.7%
- 2 a 32% b 33% c 25% d 10% e 90%
- 3

	Water late	Don't water late	Total
Mulch	10%	22%	32%
No mulch	16%	52%	68%
Total	26%	74%	100%

a 10% b 52% c 22% d 48%

- 4 a 82.4% b 22.7%
c households with mains water and a rainwater tank
e i 77% ii 17.3% iii 0.3%

18 Clothes for a cool Earth

- 1 a i B ii D iii A & B iv B & C v A & B & C
b i B (organic and living wage)
ii D (not organic, not living wage)

2 a

	Hot water	Cold water	Total
Full load	60%	25%	85%
Part load	5%	10%	15%
Total	65%	35%	100%

b i 60% ii 10% iii 65%

c i part load, hot water ii full load, cold water

- 3 a 58 b 20 c 5 d 3 e 2 f 15

- 4 b 4 c $\frac{4}{40}$ or 10% d $\frac{31}{40}$

19 Exponential growth

- 1 a 1, 2, 4, 8, 16, 32, 64, 128

Selected answers (cont'd)



b

t	0	1	2	3	4	5	6	7	8
P	1	2	4	8	16	32	64	128	256

c 1 year d 100%

2 a

t	0	1	2	3	4	5	6	7	8
P	1	1.1	1.21	1.331	1.464	1.611	1.772	1.949	2.144

b approx. 7.3 years

3 a

t	0	1	2	3
P	1	1.2	1.44	1.728
% inc.	--	20%	20%	20%

b $P = (1.2)^t$ c 2.073 6 d less than 4 years

t	1	2	3	4	5	6	7	8	9	10
P	1	4	9	16	25	36	49	64	81	100

b i less than 1 year ii more than 2 years iii 3 years
c As time goes on, it takes longer for the population to double. In exponential growth doubling time stays the same forever.

5

Country	2011 Annual growth rate (%)	Doubling time	Year in which population will be double 2011 level
Australia	1.2	58.3	2070
Brazil	0.9	77.78	2089
Canada	1.0	70	2081
China	0.5	140	2151
Germany	0.0	n/a	Never
Japan	0.3	233.3	2245
Timor-Leste	2.9	24.14	2036
United Kingdom	0.8	87.5	2099
United States	0.7	100	2111

6 a $k = \frac{70}{D}$ 70/D b 45 years c 1.56%
d i 2056 ii 2101
e Answers will vary.

20 No more bottled water

- 1 a Bright b 6b c simultaneous
d first intercepts $b = 1100$, $a = 1100$;
second intercepts $6b = 6000$, $b = 1000$
 $5a = 6000$, $a = 1200$
600 Aurora, 500 Bright
e $5a + 5b = 5500$ (3); $b = 500$, $a = 600$
600 Aurora, 500 Bright
- 2 a $10J + 100G = 250$ (1)
 $15J + 80G = 235$ (2)
b line 1 intercepts $G = 2.5$, $J = 25$
line 2 intercepts $G = 2.9375$, $J = 15.67$
lines cross at $J = 5$, $G = 2$
c $30J + 300G = 750$; $30J + 160G = 470$;
take (4) from (3): $140G = 280$, $G = 2$; $J = 5$
i.e. a jug costs \$5 and a glass costs \$2
- 3 a $20A + 50P = 1300$; $10A + 100P = 1400$
c $A = \$40$, $P = \$10$

21 Earth - the only home we have

- 1 a i 42 000 000 ii 4.2×10^7
b i 1 000 000 000 ii 1×10^9 c i and iii
- 2 a 4 600 000 000 b 4.6×10^9 c six d 4.5×10^9
e i one hundred million years ii 100 000 000

iii 1×10^8

- 3 a 2.42×10^9 , 5.41×10^8
b i 1 879 000 000 ii 1.879×10^9
c 6.2×10^8 years d 4.59×10^8 years
- 4 a 1.5×10^6 b 1×10^9 c 1.5×10^{15} m³
- 5 a 39 800 000 b 3.98×10^7
- 6 a $3.153 6 \times 10^7$ b $9.460 8 \times 10^{12}$
c 2.081 337 6 $\times 10^{14}$ km d ii 38 million
e 19 million years f No surface water, no air

Acknowledgements

p13: Australian Bureau of Statistics 2009-10, *Waste Management Services, Australia 2009-10*, 'Table 16: Waste streams of material received at landfills, by source', data cube: Excel spreadsheet, cat. no. 8698.0, viewed 30 September 2013, <<http://www.abs.gov.au/AUSSTATS/abs@.nsf/Lookup/8698.0Main+Features12009-10?OpenDocument>>

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p.34 Australian Bureau of Statistics 2011, *Land Account: Great Barrier Reef regions, Experimental Estimates, 2011*, 'Table 5.6: Vegetation cover 2006 and pre-1750 vegetation type, Total GBR region, 2009', data cube: Excel spreadsheet, cat. no. 4609.0.55.001, viewed 30 September 2013, <<http://www.abs.gov.au/AUSSTATS/abs@.nsf/Lookup/4609.0.55.001Main+Features12011?OpenDocument>>

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p.37 Australian Bureau of Statistics 2010, *Environmental issues: Water use and conservation, March 2010*, 'Table 2: Sources of water for households - 1998 to 2010', data cube: Excel spreadsheet, cat. no. 4602.0.55.003, viewed 30 September 2013, <<http://www.abs.gov.au/AUSSTATS/abs@.nsf/DetailsPage/4602.0.55.003Mar%202010?OpenDocument>>

p.41: The World Bank: Population growth (annual %): Derived from total population. Population source: (1) United Nations Population Division. World Population Prospects, (2) United Nations Statistical Division. Population and Vital Statistics Report (various years), (3) Census reports and other statistical publications from national statistical offices, (4) Eurostat: Demographic Statistics, (5) Secretariat of the Pacific Community: Statistics and Demography Programme, and (6) U.S. Census Bureau: International Database. See terms of use: <http://go.worldbank.org/OJC02YMLA0>

Selected answers



1 Solar power

- 1 a 10 b 30
2 2 kW
3 a 1 653 300 mm² b 1.67 m by 0.9 m c 1.653 3 m²
d 1 000 000 (one million)
4 a 1.98 m² b 19.8 m² c 240 m² d 8.25% e 27
f 108
5 a 18 b 3 c 54 d 1.264 m² e 68.256 m²
f 6.744 m²
6 a 16 m² b 6.48 m² c 40.5% d 2.7 m e 5.4 m
f 7.29 m²

2 Where the wind blows

- 1 a 20 000 b 1250 c 4
2 $\sin x = \frac{O}{H}$, $\cos x = \frac{A}{H}$, $\tan x = \frac{O}{A}$
3 70 m b i 107 m ii 37 m
4 a 64° b 185 m c 206 m
5 a 58° b 26 m
6 a 36 m b 137 m c 67°

3 Biogas power

- 1 a 0.8 m b 1.152 m³ c i 1.44 m² ii 0.96 m²
d 5.28 m²
2 a 1.75 m b 11.225 m³ c 0.55 m d 0.855 m³
e 12.08 m³ f 12 080 L g 3 020 L
3 a 1.5 m b 0.825 m³ c 825 L d 9.7%
4

Middle	cylinder	$L = 5$ m $d = 2.4$ m $r = 1.2$ m	22.62 m ³	37.70 m ²
End	hemisphere	$d = 2.4$ m $r = 1.2$ m	3.62 m ³	9.05 m ²

b i 29.86 m³ ii 55.8 m²

4 Forests forever

- 1 a 5 100 mm² b 0.005 1 m² c 0.025 5 m³
d 0.002 5 m² e 0.012 5 m³ f 0.267 m³
2 a i 0.15 m ii 0.071 m² b 0.355 m³ c 0.088 m³
3 a i 0.003 8 m² ii 0.022 8 m³ b 0.182 4 m³
c 0.030 6 m³ d 0.483 m³ e 0.255 m³ f 52.8%
4 a i 0.481 m³ ii 5.498 m² b 0.027 5 m³ c 5.7%
5 a 17.6% b 17.1%

5 A load of rubbish

- 1 a 6240 litres b 6.24 m³ c 6 years
2 167 bins
3 a 7.656 9 × 10⁶ km² b 0.5%
4 a i 10 145 t ii 10 812 t
b i 1.014 5 × 10⁴ t ii 1.081 2 × 10⁴ t
c 6.67 × 10² t
d i 3.39 × 10³ t ii 3.622 × 10³ t
e i 3.39 × 10⁶ kg ii 3.622 × 10⁶ kg
f i 6.87 × 10² t ii 6.87 × 10⁵ kg g 3.5 × 10⁻²
h 1.64 × 10⁻¹
5 a 8.91 × 10⁵ b i 6 534 000 ii 6.534 × 10⁶
c i 21 626 000 ii 2.162 6 × 10⁷ d 22 e 8.11 × 10⁵

6 Recycling plastic

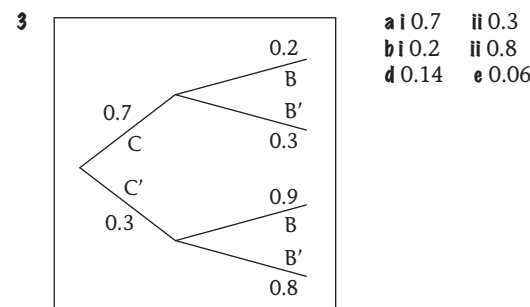
- 1 a i 16 ii 32 b 16 c 19 d 22 e 24
2 a 1 8 b 29 c 31 d 18
2 2 3 4
2 6 7 8 9
3 1 1 2 3 4
3 5 6
3 a 3 4 b i 45 ii 45 iii 18
3 8 9 c Yes, the median has increased from
4 2 3 4 22 to 45 and mode from 22 to 45.
4 5 5 5 6 7 8 The average number of bottles
5 0 1 2 recycled has more than doubled.
4 a 10 5 9 b i 126.5 ii 127 iii 45
11 2 4 5 6 6 7 8 9 c 10
12 2 3 4 5 6 7 7 7 9
13 0 1 4 5 6 8
14 0 1 6
15 0
5 a i 27 ii 26 iii 26 b i 20 ii 15 iii 15 c Bin B
d Discuss with teacher.

7 Let it rot

- 1 a 30 b 60:1
2 a paper, straw b grass cuttings, weeds, food waste, cow
manure c carbon, nitrogen d straw e food waste
3 a i 3:2 ii 2:1 iii 9:10 iv 4:5 v 4:3 b iii
4 a 2 kg b 4 kg c 0.5 kg d 2.5 kg
5 a 6 kg b 55 kg
6 a 20 kg b 10 kg
7 a 400 kg b 240 kg
8 a 1200 kg b 800 kg

8 On yer bike

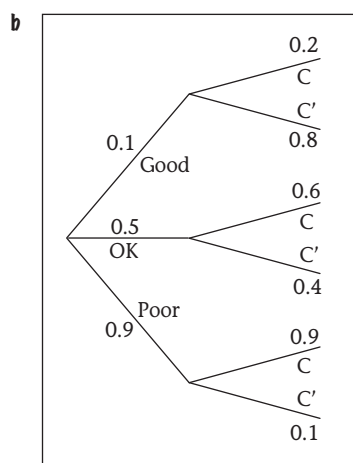
- 1 4WD, small car, bus, bike
2 a owns car but no bike b doesn't own car, owns bike
c doesn't own car or bike



- 4 a cool weather, Grace doesn't ride to work b 0.18
c 0.36 d 0.54
5

Quality	P(quality)	P(car owner)
Good	0.1	0.2
OK	0.5	0.6
Poor	0.4	0.9

Selected answers (cont'd)



c i 0.08 ii 0.04 iii 0.32

9 Flying carbon neutral

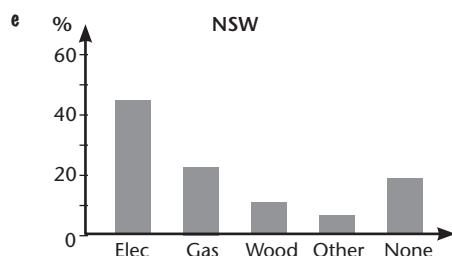
- a i \$18 ii \$54 iii \$7.20 b i 10 t ii 50 t
- a i short ii medium
- Discuss with teacher.
- a 0.4 b 294 kg c 0.294 t
- a 0.25 b 1578.5 kg c 1.578 5 t d \$31.57
- a 350 kg b 1750 km
- long
-

Seat class	CO ₂ per passenger
Economy	0.18
Business	0.38
First	0.57

- a 1.5 b 3.17
- a 5.7t b i 15 000 km ii 31 667 km
- a \$70 b \$20 c D = 20C
- a 4 t b \$22 c D = 22C

10 Comfort zone

- a i Electricity ii Gas b Vic, WA, ACT
c i 13.7 ii 98.1 d A: Vic., B: WA



- a 37.1 b 38.6 c 65.6, 7.7 d 57.9
e i 37.3 ii 38.6 f outliers
- a 2, 2, 4, 7, 15, 19, 48, 86 b i 22.9% ii 11 iii 2
c median
- a i 541,000 ii 606,400 b i 68.8 ii 86.3
c i NT ii Tas d i 60.44 ii 74.8 e i 59.0 ii 74.6

11 Don't let the tap run dry

- a $t \leq 4$ b $V > 500$ c $t > 4$ d $V \leq 500$
- c
- a $t \leq 5$ b $8.5t \leq 60$ c $t \leq 7$

- a $20t + 15 \leq 300$ b $12N + 25 > 200$
- c, i
- a 0.002 5V b $0.002\ 5V + 28 < 300$ c 108 800 L
d 108.8 kL
- a 29 500 L b $V \leq 29\ 500$ c $90d < 29\ 500$
d $90d + 360 < 29\ 500$ e $d < 324\ L$

12 Food miles

- a (0, 0), origin b E = 300, 300 g
c D = 480, E = 1200, 480 km, 1200 g d 2.5
e D, E, 2.5, 0 f E = 2.5D g 8 250 g h 400 km
- a (0, 500), intercept b 2.5 c E = 2.5D + 500
d 7 250 g
- a (0, 0.4) b E = 1.15, 1.15 kg c (3000, 2.65),
D = 3000, E = 2.65, 3000, 2.65 d 0.000 75
e E = 0.000 75D + 0.4 f 5.95 kg
- a Line B b Line A c Line D d Line C
- a 350 b -0.2 c S = -0.2D + 350 d 1 750 km

13 The price of meat

- a 4 164 000 t b 78 847 000 t c 42 551 000 d 31%
e 1.46%
- a i India ii USA b i 9.5 kg ii 21.3 kg
c i 8.1% ii 55.8% d Australia e Kenya
- a 41 000 000 ha b i 1.2% ii 0.6%
c 131 million ha d 3.2% e 1233 million ha
- a 492.8 kg b 42% c 235.2 kg

14 Vegie gardens

- a 7.5 kg b 37.5 kg c 17 kg d i 30 kg ii 4
- a

Rows	1	2	3	4	5
Seedlings	8	16	24	32	40

b i 64 ii 120 c 0.25 kg d 3 kg e i 500 L ii 71 L

- a 20 m² b 120 c i 0.4 bale ii 8 bales
iii 2.67 or $2\frac{2}{3}$ bales d 4 bags e 36 f 16 kg
- a 0 b 15 cm/week c 30 cm
d i 0.4 kg ii 0.067 kg iii 150
- a Graph D b Graph A c Graph B

15 Electric pedal power

- a \$200 b 10 c \$53.85
- electric – Line A, non-powered – Line B
- a i 40 min ii 24 min b i 16 kmh⁻¹ ii 9.6 kmh⁻¹
- 60
- a 3 h 12 min b 1 h 12 min
- a Graph C b Graph D c Graph A
-

Section	AB	BC	CD	DE	EF	FG	GH	HI
Statement	c	b	d	e	a	c	d	b

16 We all need trees

- Eucalypt woodlands
- Low, closed forests
- Cleared, non-native, buildings; freshwater, salt lakes, lagoons; regrowth, modified native vegetation
- a Eucalypt woodlands
b Cleared, non-native, buildings
- a Eucalypt woodlands, eucalypt forests, acacia forests

Selected answers (cont'd)



- and woodlands
b Cleared, non-native, buildings, eucalypt woodlands, eucalypt forest
- 6 Low, closed forests, heathlands, sea and estuaries
- 7 a 7 661 ha b 14 193 ha
- 8 a forests, open woodlands, woodlands
b i 18 120 037 ii 28 110 716
c i 21.7% ii 20.6% d i 58.3% ii 15.3%

9 a

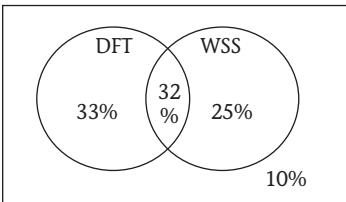
Type	2006 area	1750 area	Change
Acacia	1 331 707	5 279 798	3 948 091
Callitris	163 436	225 752	62 316
Casuarina	54 654	69 872	15 218
Melaleuca	234 261	328 292	94 031
Other	165 445	174 456	9 011

- b i Acacia ii Other c 4 128 667 ha
- 10 a 1 261 064 ha b 9 990 679 ha
- 11 15 380 410 ha

17 Precious water

- 1 a 12.6% b 26.7% c 12.41% d 2.82% e 63.1%
f 2.30% g 45.7%

2



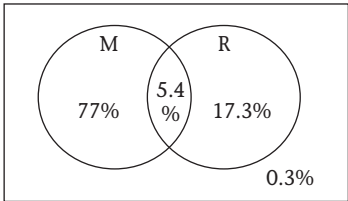
- a 32% b 33% c 25% d 10% e 90%

3

	Water late	Don't water late	Total
Mulch	10%	22%	32%
No mulch	16%	52%	68%
Total	26%	74%	100%

- a 10% b 52% c 22% d 48%
- 4 a 82.4% b 22.7%
c households with mains water and a rainwater tank

d



- e i 77% ii 17.3% iii 0.3%

18 Clothes for a cool Earth

- 1 a i B ii D iii A & B iv B & C v A & B & C
b i B (organic and living wage)
ii D (not organic, not living wage)

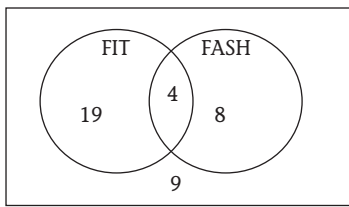
2 a

	Hot water	Cold water	Total
Full load	60%	25%	85%
Part load	5%	10%	15%
Total	65%	35%	100%

- b i 60% ii 10% iii 65%
c i part load, hot water ii full load, cold water

3 a 58 b 20 c 5 d 3 e 2 f 15

4 a



- b 4 c $\frac{4}{40}$ or 10% d $\frac{31}{40}$

19 Exponential growth

- 1 a 1, 2, 4, 8, 16, 32, 64, 128

b

t	0	1	2	3	4	5	6	7	8
P	1	2	4	8	16	32	64	128	256

- c 1 year d 100%

2 a

t	0	1	2	3	4	5	6	7	8
P	1	1.1	1.21	1.331	1.464	1.611	1.772	1.949	2.144

- b approx. 7.3 years

3 a

t	0	1	2	3
P	1	1.2	1.44	1.728
% inc.	--	20%	20%	20%

- b $P = (1.2)^t$ c 2.073 6 d less than 4 years

t	1	2	3	4	5	6	7	8	9	10
P	1	4	9	16	25	36	49	64	81	100

- b i less than 1 year ii more than 2 years iii 3 years
c As time goes on, it takes longer for the population to double. In exponential growth doubling time stays the same forever.

5

Country	2011 Annual growth rate (%)	Doubling time	Year in which population will be double 2011 level
Australia	1.2	58.3	2070
Brazil	0.9	77.78	2089
Canada	1.0	70	2081
China	0.5	140	2151
Germany	0.0	n/a	Never
Japan	0.3	233.3	2245
Timor-Leste	2.9	24.14	2036
United Kingdom	0.8	87.5	2099
United States	0.7	100	2111

- 6 a $k = \frac{70}{D}$ 70/D b 45 years c 1.56%

- d i 2056 ii 2101
e Answers will vary,

20 No more bottled water

- 1 a Bright b 6b c simultaneous
d first intercepts $b = 1100$, $a = 1100$;
second intercepts $6b = 6000$, $b = 1000$
 $5a = 6000$, $a = 1200$
600 Aurora, 500 Bright

Selected answers (cont'd)



e $5a + 5b = 5500$ (3); $b = 500$, $a = 600$
600 Aurora, 500 Bright

- 2 a $10J + 100G = 250$ (1)
 $15J + 80G = 235$ (2)
b line 1 intercepts $G = 2.5$, $J = 25$
line 2 intercepts $G = 2.9375$, $J = 15.67$
lines cross at $J = 5$, $G = 2$
c $30J + 300G = 750$; $30J + 160G = 470$;
take (4) from (3): $140G = 280$, $G = 2$; $J = 5$
i.e. a jug costs \$5 and a glass costs \$2
- 3 a $20A + 50P = 1300$; $10A + 100P = 1400$
c $A = \$40$, $P = \$10$

21 Earth - the only home we have

- 1 a i 42 000 000 ii 4.2×10^7
b i 1 000 000 000 ii 1×10^9 c i and iii
- 2 a 4 600 000 000 b 4.6×10^9 c six d 4.5×10^9
e i one hundred million years ii 100 000 000
iii 1×10^8
- 3 a 2.42×10^9 , 5.41×10^8
b i 1 879 000 000 ii 1.879×10^9
c 6.2×10^8 years d 4.59×10^8 years
- 4 a 1.5×10^6 b 1×10^9 c $1.5 \times 10^{15} \text{ m}^3$
- 5 a 39 800 000 b 3.98×10^7
- 6 a 3.1536×10^7 b 9.4608×10^{12}
c $2.0813376 \times 10^{14} \text{ km}$ d ii 38 million
e 19 million years f No surface water, no air

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